maelcum mixer — readable schematics of the generated boards

One page per board. Each drawing is generated from the same netlist as the board's PCB
and checked against it in both directions (pcb/tools/leggibile.py). Values are PROVISIONAL drafts.

Page	Board
2	maelcum CHANNEL CDJ/LINE (v0.4) — draft v0.1 (383 parts, rev. 0.2-draft)
3	maelcum FX-RET (dec. 130) — draft v0.1 (199 parts, rev. 0.1-draft)
4	maelcum FX-RET reroute (dec. 91) — draft v0.1 (225 parts, rev. 0.1-draft)
5	maelcum FX-SEND (dec. 91a) — draft v0.2 (120 parts, rev. 0.3-draft)
6	maelcum FX-SEND control board — draft v0.2 (4 parts, rev. 0.2-draft)
7	maelcum MASTER signal board (dec. 140) — draft v0.1 (285 parts, rev. 0.2-draft)
8	maelcum MASTER OUTPUT board L (dec. 142-143) — draft v0.1 (155 parts, rev. 0.1-draft)
9	maelcum MASTER OUTPUT board R (dec. 142-143) — draft v0.1 (144 parts, rev. 0.1-draft)
10	maelcum MASTER control board — draft v0.1 (60 parts, rev. 0.1-draft)
11	maelcum MIC (dec. 97) — draft v0.1 (213 parts, rev. 0.1-draft)
12	maelcum MIC control board — draft v0.1 (10 parts, rev. 0.1-draft)
13	maelcum MONO (dec. 97) — draft v0.1 (127 parts, rev. 0.1-draft)
14	maelcum MONO control board — draft v0.1 (9 parts, rev. 0.1-draft)
15	maelcum PSU-IN (dec. 113) — draft v0.1 (41 parts, rev. 0.1-draft)

How to read the drawings

- Arrow and ground symbols are power rails and grounds; the name under the symbol is the rail (AGND, +18V, −15SW ...).
- Resistors and capacitors are wired to the op-amp pin they touch: in series along the pin, shunt towards the rail.
- A plain name on a wire is a net that continues elsewhere in the same block.
- A name in a box is a net that goes to another block on the same board, or to a connector.
- Connectors and many-pin chips (C96, LM339, ADG1434) carry only names; their parts are drawn in chains below them.
- Dashed frames are the functional blocks of the board, in signal order. Net names are the PCB identifiers.

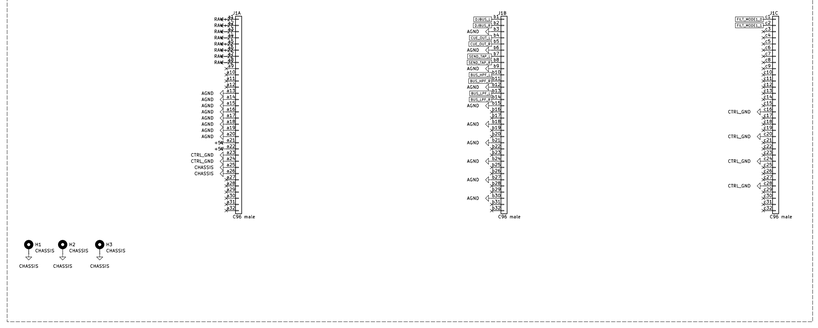
Generated by pcb/tools/leggibile.py

Sheet: /
File: schematics-cover.kicad_sch

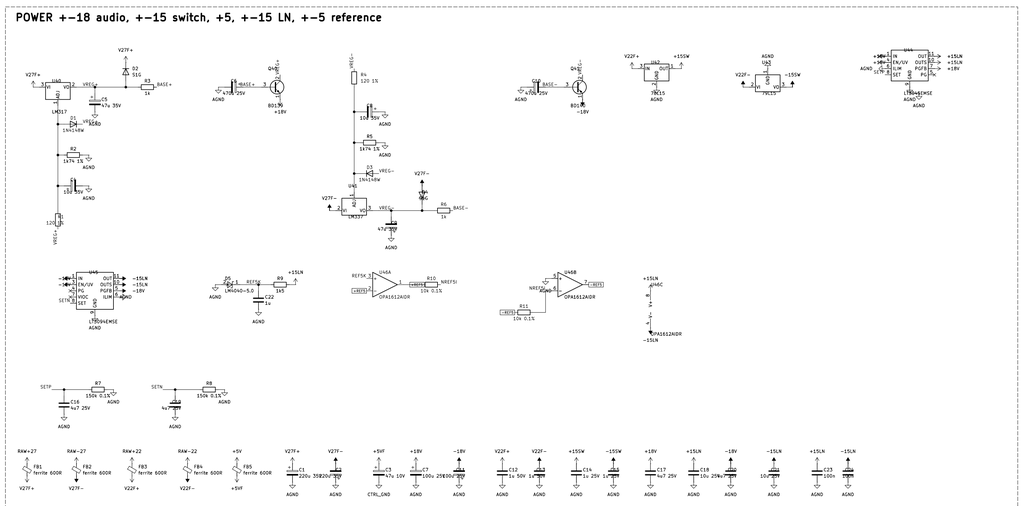
Title: maelcum mixer — readable schematics

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 1/1

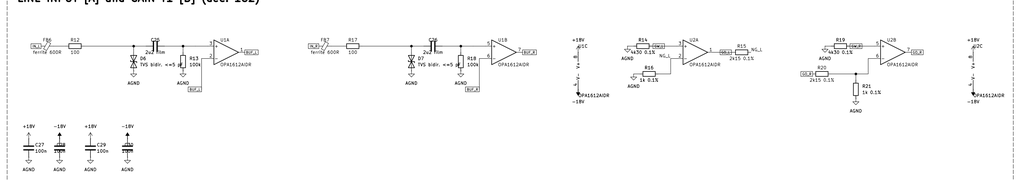
BACKPLANE C96 (CDJ class, rev. dec. 159)



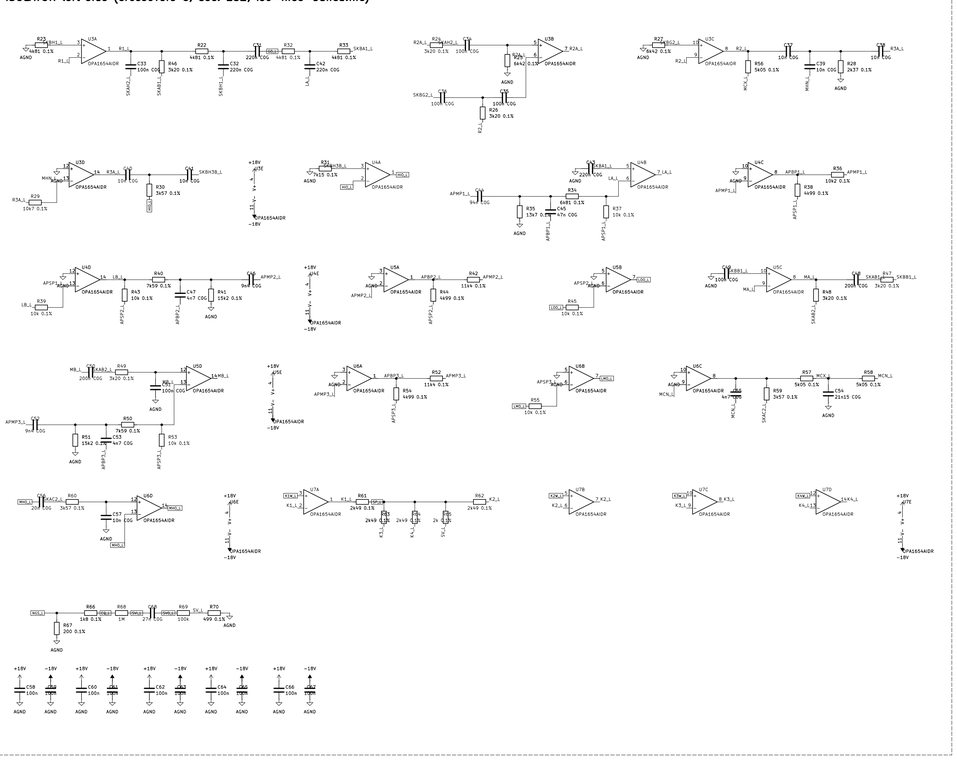
POWER +−18 audio, +−15 switch, +5, +−15 LN, +−5 reference



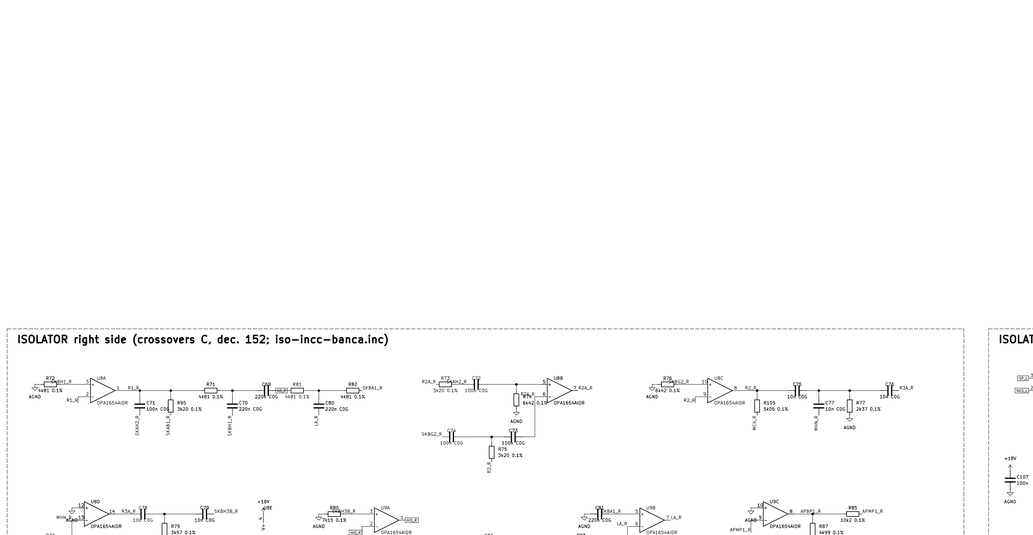
LINE INPUT [A] and GAIN T1 [B] (dec. 162)



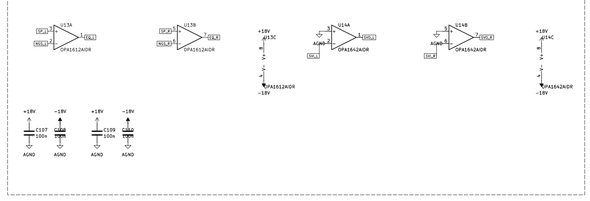
ISOLATOR left side (crossovers C, dec. 152; iso−incc−banca.inc)



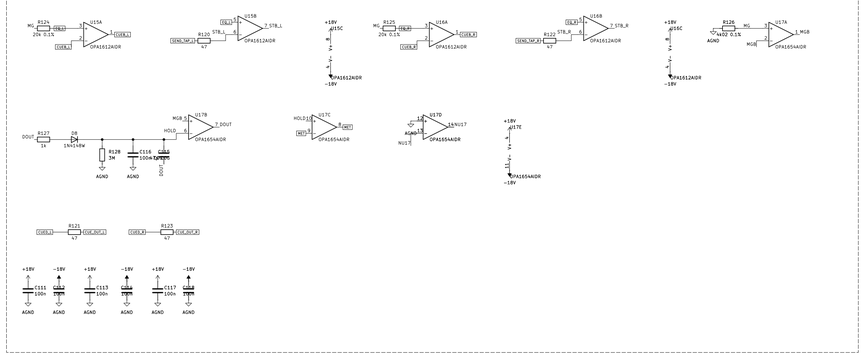
FADER [H] (2 VCAs, fader shaper, dec. 126) and DIP (dec. 163)



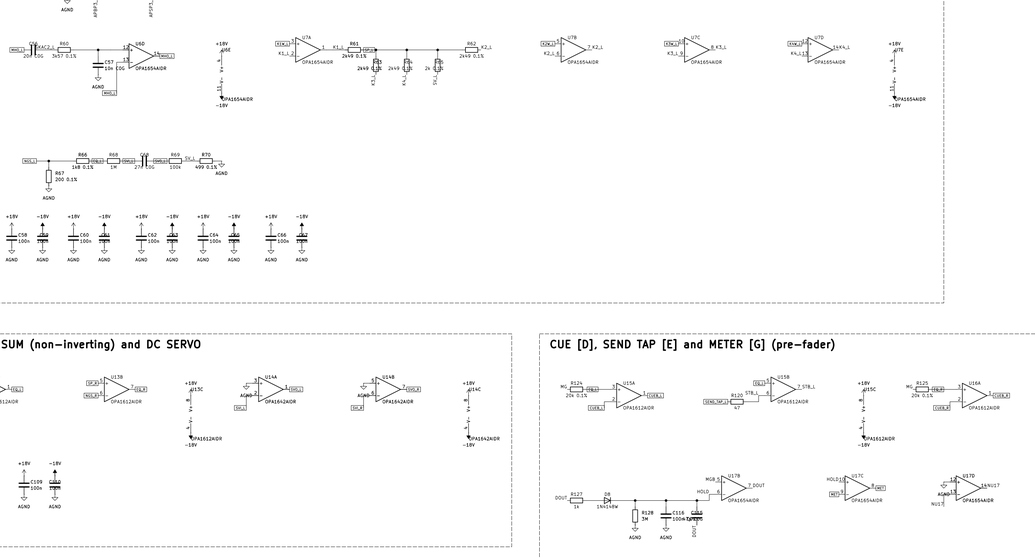
ISOLATOR SUM (non−inverting) and DC SERVO



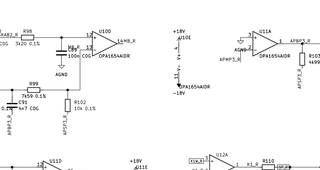
CUE [D], SEND TAP [E] and METER [G] (pre−fader)



HPF/OFF/LPF TOGGLE: self−timed logic (74HC14, 74HC86, 74HC74) and ADG1434



CONTROL BOARD CABLE (J2) and RCA INPUTS (J3)



PWR_FLAG (for the ERC check only)



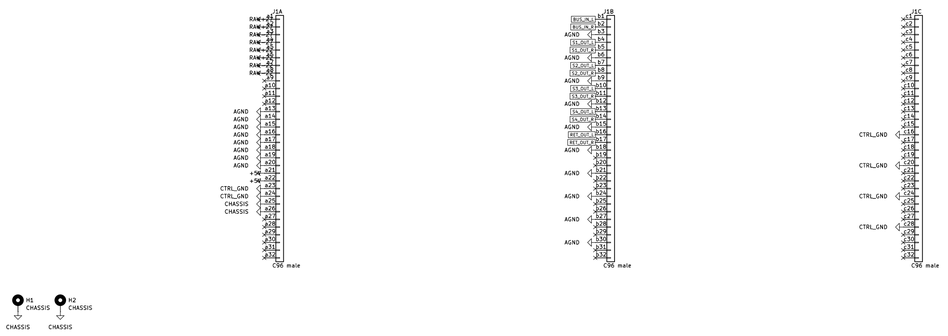
Generated by pcb/tools/leggible.py from the same netlist as the PCB; do not edit by hand

Sheet: /
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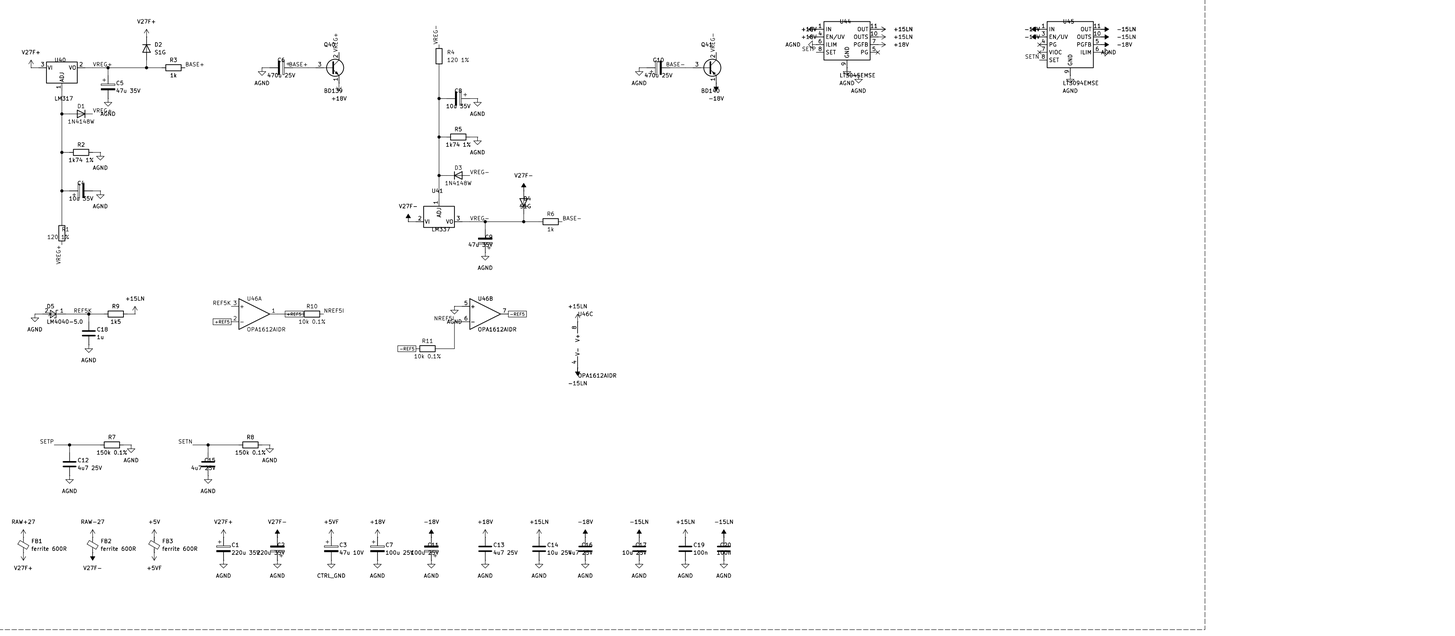
Title: maelcum CHANNEL CDJ/LINE (v0.4) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.2−draft
KiCad E.D.A. 10.0.7		Id: 1/1

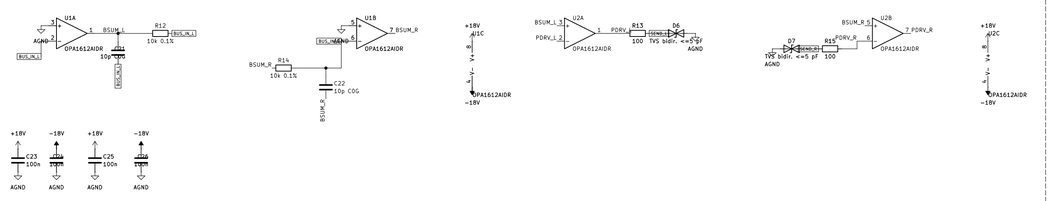
BACKPLANE C96 (FX-RET class, dec. 130)



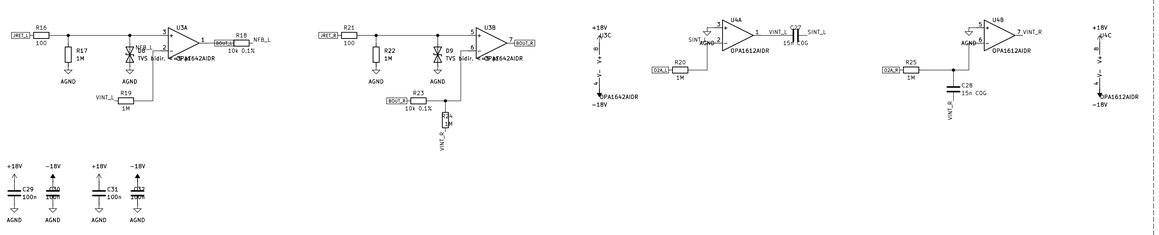
POWER +-18 audio, +5, +-15 LN, +-5 reference



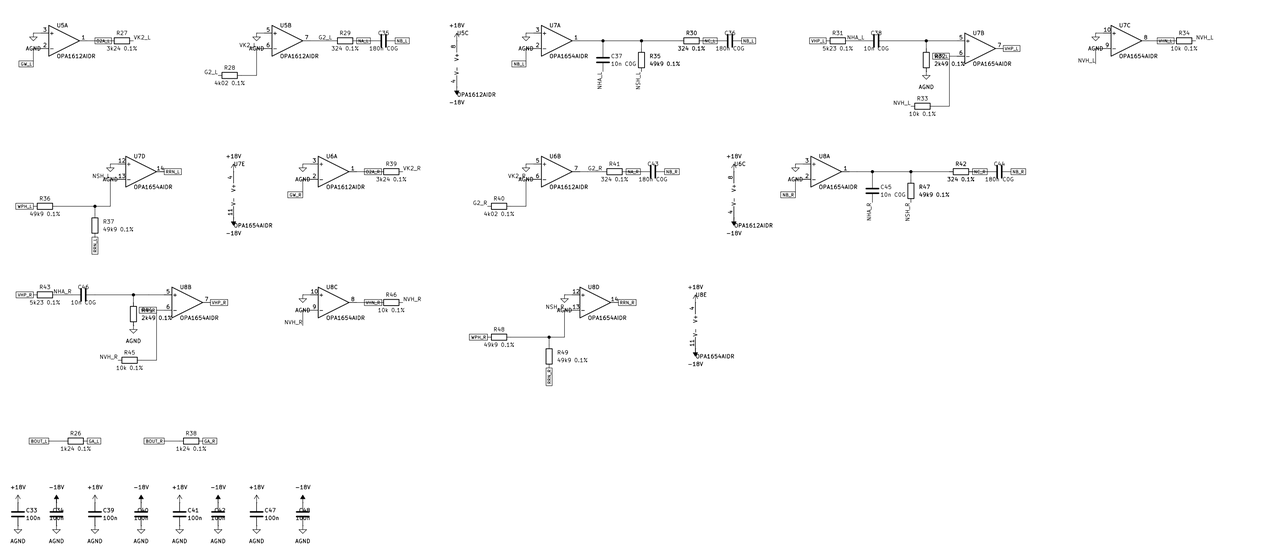
BUS SUM (slot 1, Rf 10k, Cf 10p) and PEDAL DRIVER



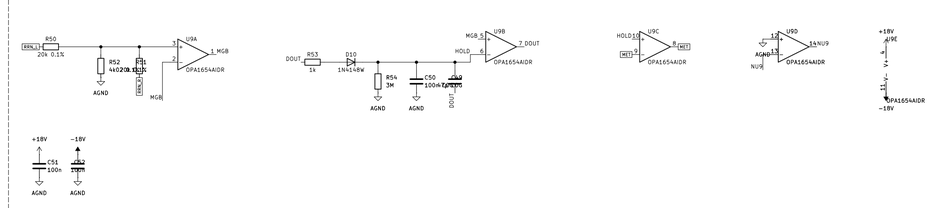
RETURN: protection, JFET buffer g=1, DC servo (ritorno-buffer-g1.cir)



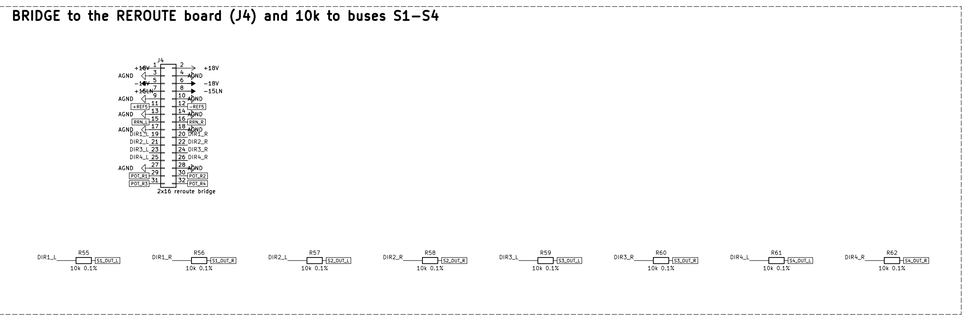
GAIN T2 (dual 10k lin pot on the cable, dec. 162; x1.24 todo 220) and EQ2



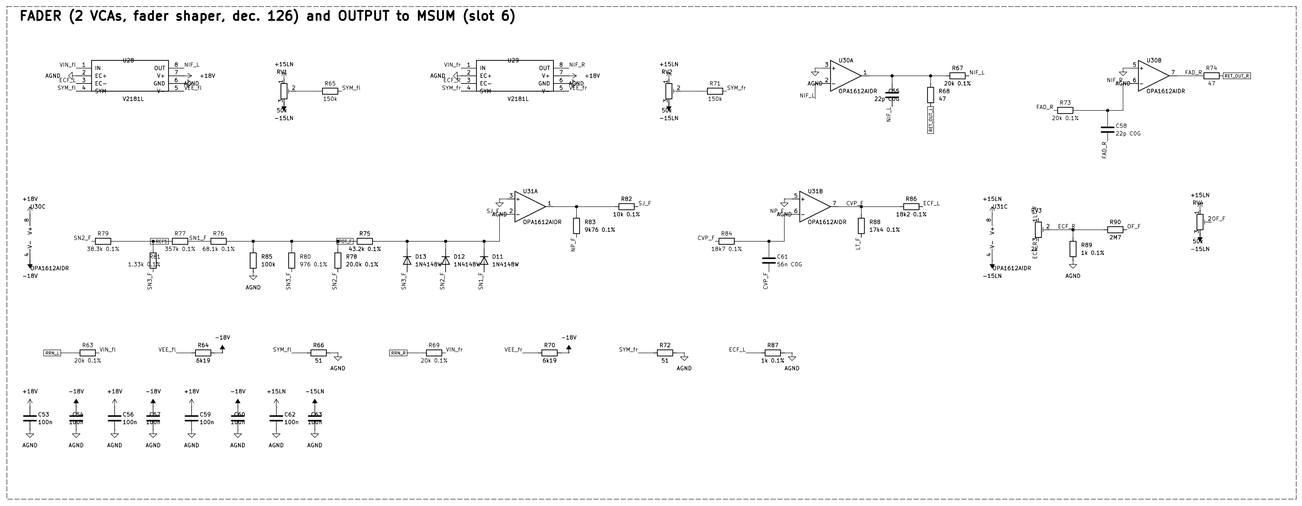
RETURN METER (centred mono, g = 0.288; detector meter-det-03s.cir)



BRIDGE to the REROUTE board (J4) and 10k to buses S1-S4



FADER (2 VCAs, fader shaper, dec. 126) and OUTPUT to MSUM (slot 6)



CONTROL BOARD CABLE (J2) and PEDAL JACK (J3)



PWR_FLAG (for the ERC check only)



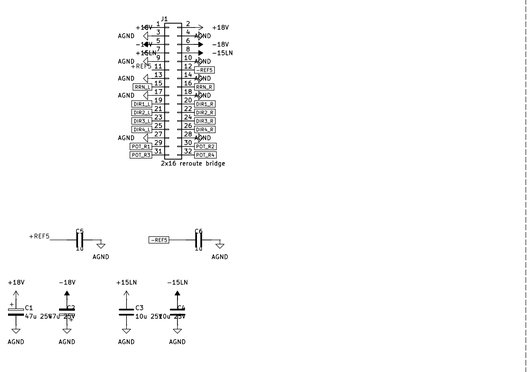
Generated by pcb/tools/leggible.py from the same netlist as the PCB: do not edit by hand.

Sheet: /
File: fx-ret.kicad_sch

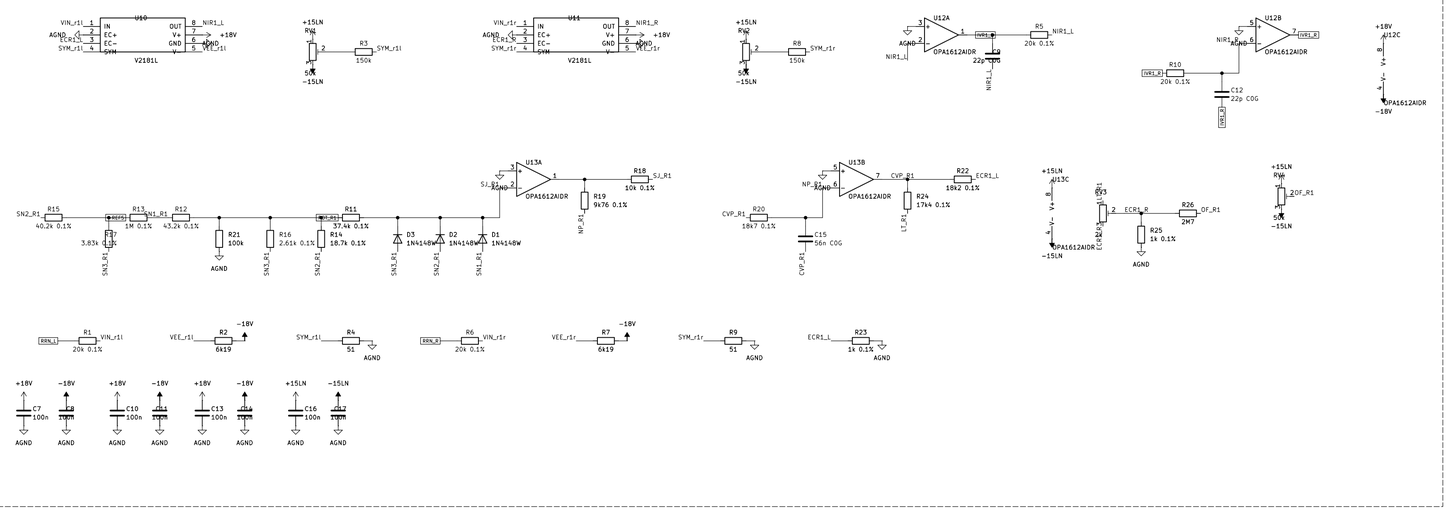
Title: maelcum FX-RET (dec. 130) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

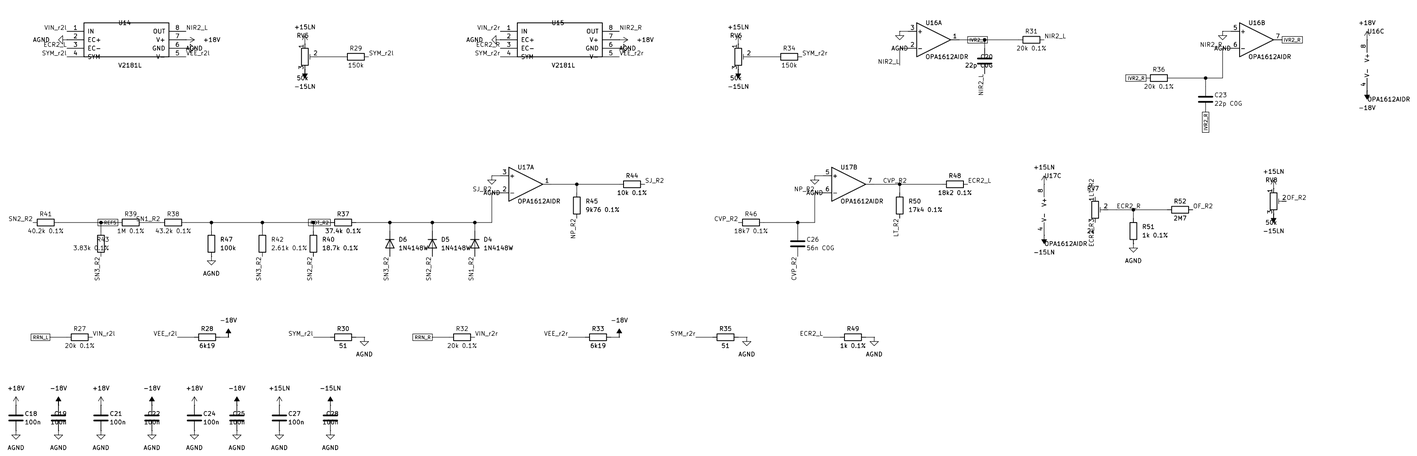
BRIDGE from the signal board (J1) and local bulk caps



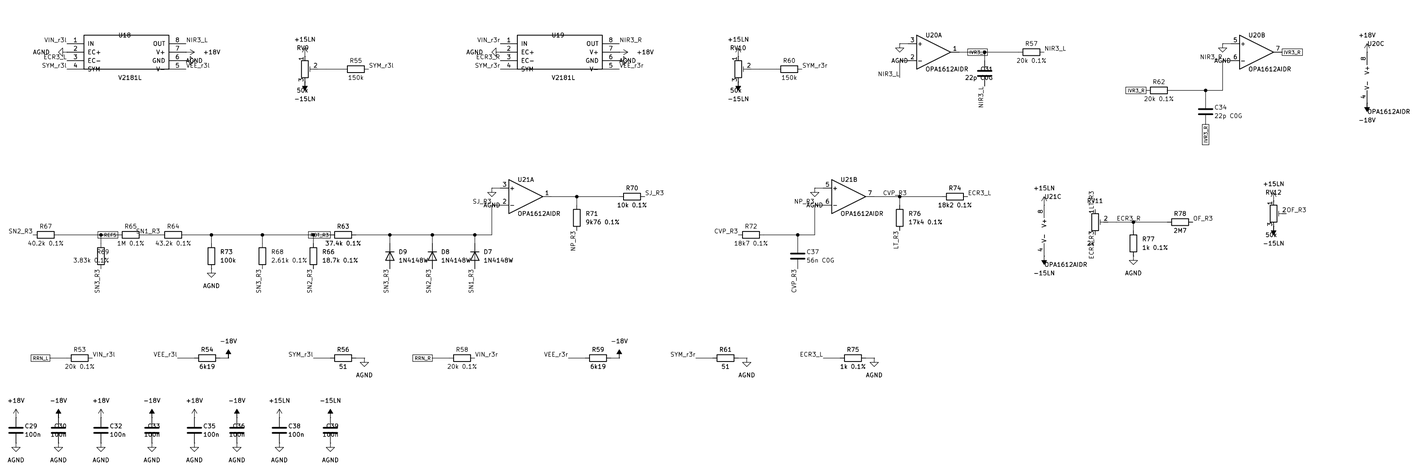
REROUTE to S1 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



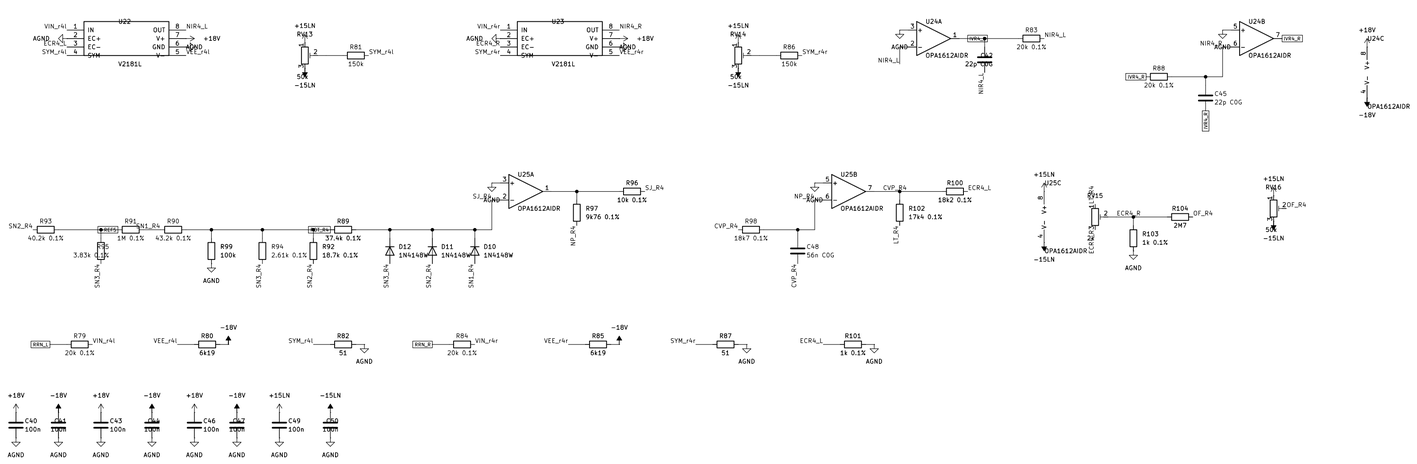
REROUTE to S2 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



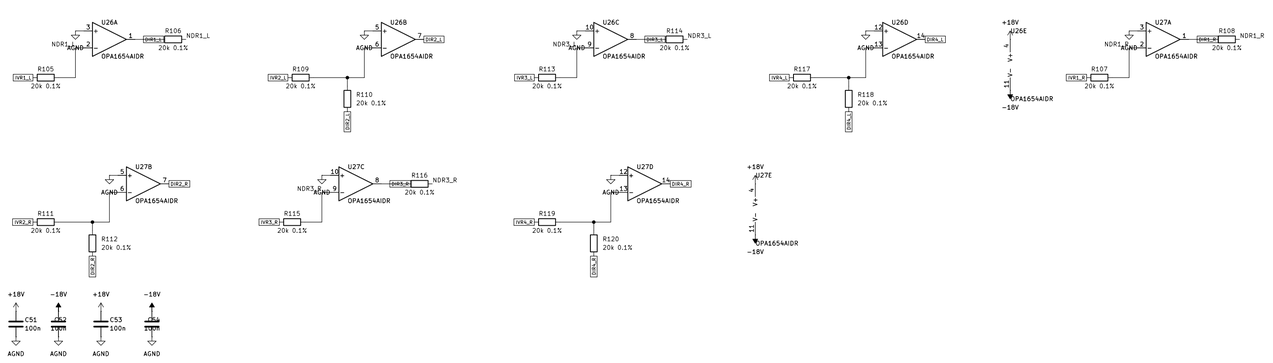
REROUTE to S3 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



REROUTE to S4 (2 VCAs, I/V, inverting driver, reroute shaper, trim)



REROUTE INJECTION DRIVERS (2 OPA1654, inverting unity)



PWR_FLAG (for the ERC check only)



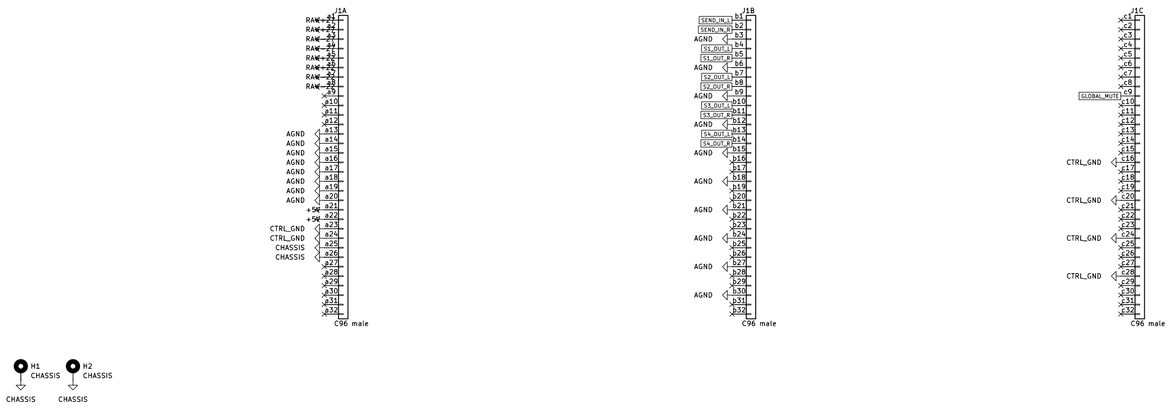
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand.

Sheet: /
File: fx-ret-reroute.kicad_sch

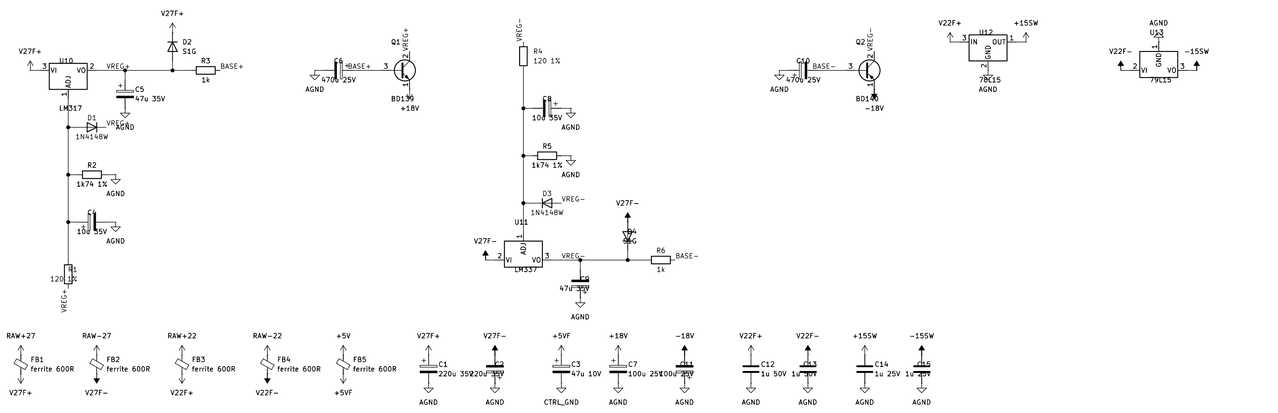
Title: maelcum FX-RET reroute (dec. 91) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

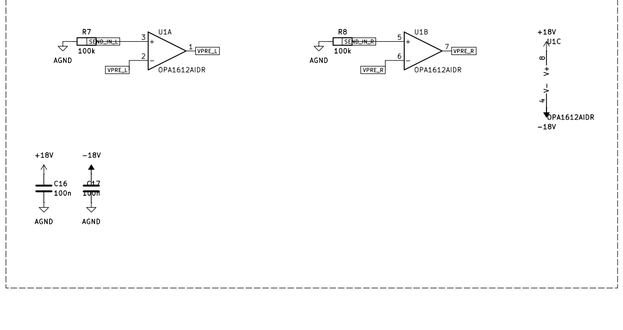
BACKPLANE C96 (FX-SEND class)



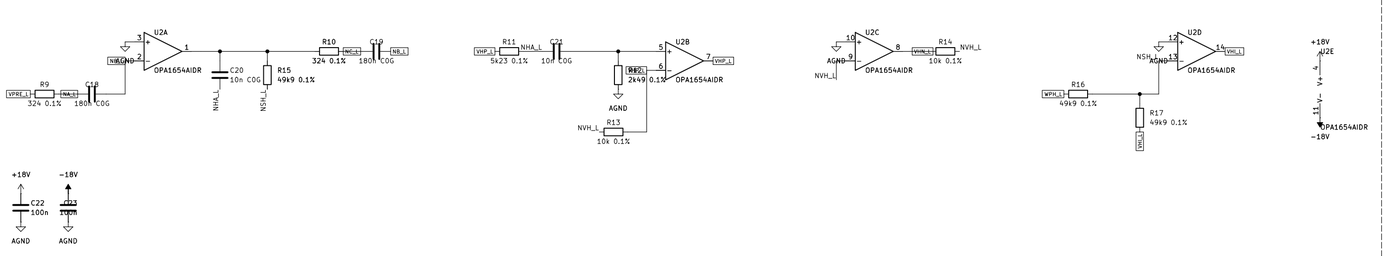
POWER +-18 audio, +-15 switch, +5



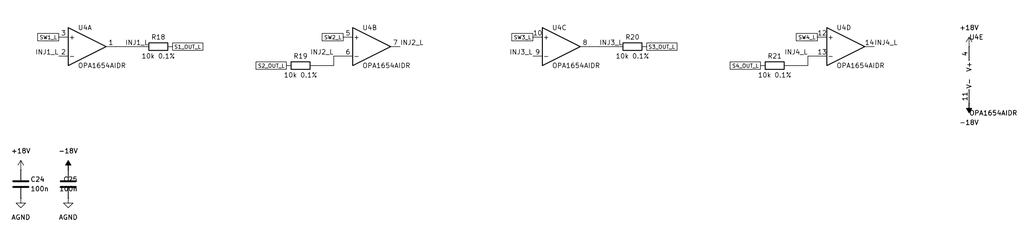
INPUT BUFFER (OPA1612)



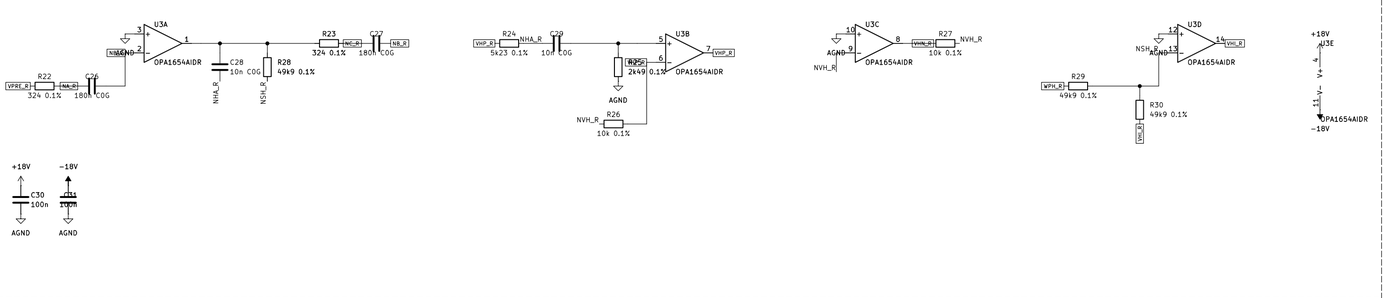
EQ2 left side (LOW 120 Hz + HIGH 4.5 kHz, fx-eq2-pot10k.cir)



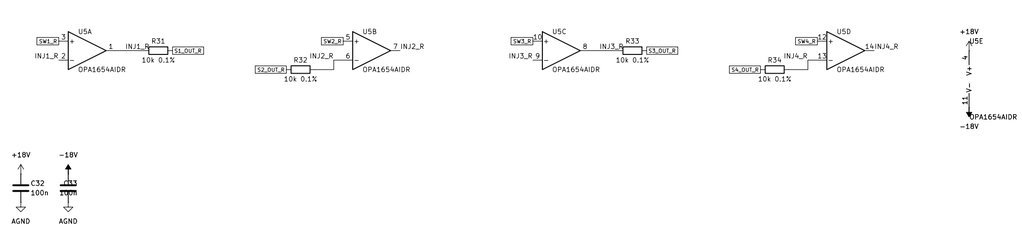
INJECTION left side (OPA1654 + 10k to the buses)



EQ2 right side (LOW 120 Hz + HIGH 4.5 kHz, fx-eq2-pot10k.cir)



INJECTION right side (OPA1654 + 10k to the buses)



ASSIGN S1-S4 (ADG1434 pi SPDT, 1 ms soft switch, global mute)

CONTROL BOARD CABLE (J2) and ASSIGN BUTTONS (J3)

PWR_FLAG (for the ERC check only)

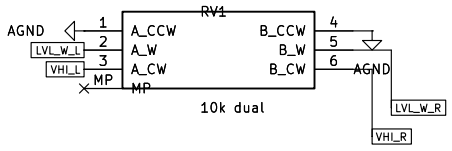
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

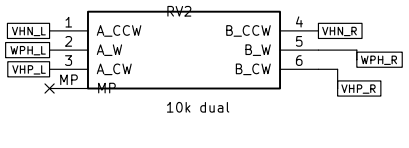
Sheet: /
File: fx-send.kicad_sch

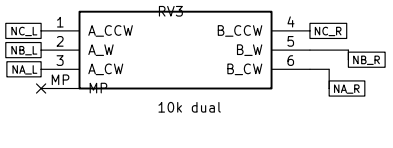
Title: maelcum FX-SEND (dec. 91a) — draft v0.2 — readable schematic

Size: User | Date: | Rev: 0.3-draft
KiCad E.D.A. 10.0.7 | Id: 1/1

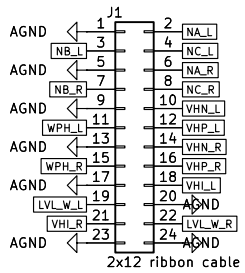
POTS (positions from the panel render)



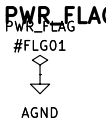




CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)



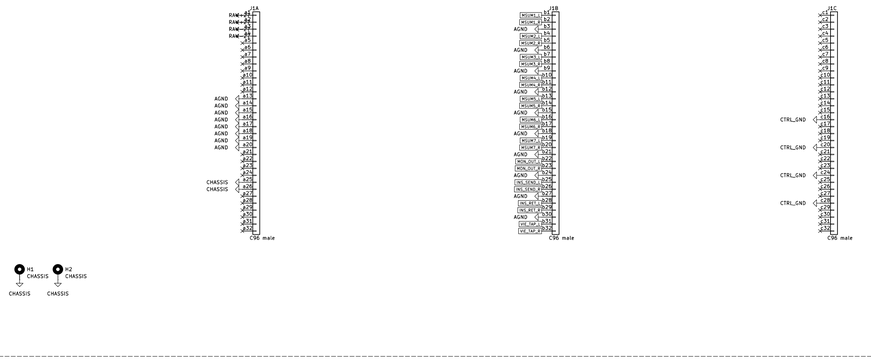
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: fx-send-comandi.kicad_sch

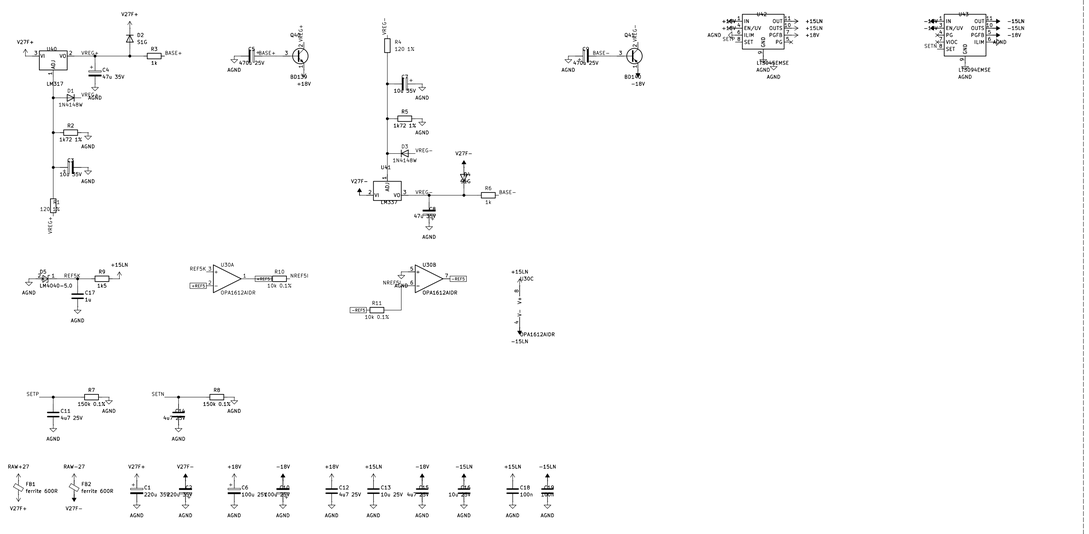
Title: maelcum FX-SEND control board — draft v0.2 — readable schematic

Size: User	Date:	**Rev: 0.2-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

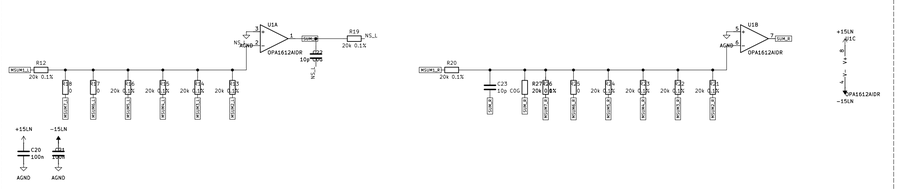
BACKPLANE C96 (MASTER class)



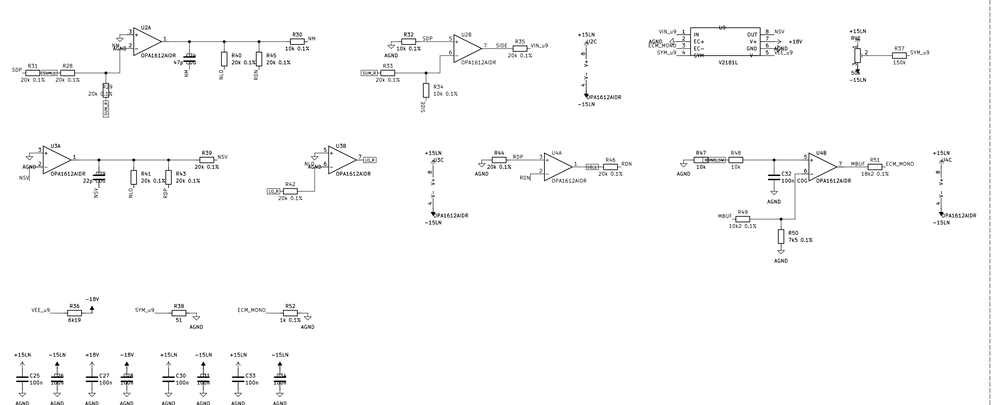
POWER +−18 (LM317/337 + capacitance multiplier), +−15 LN (LT3045/LT3094)



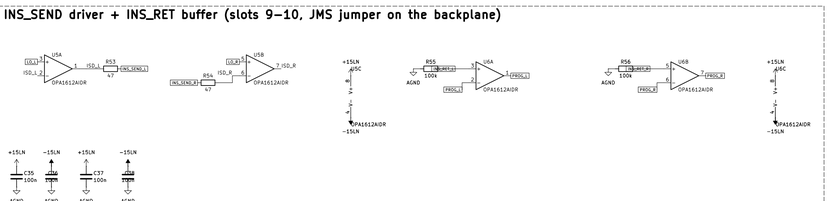
MASTER SUMMING [A2] (inverting, 20k/20k/10p, sum−noise−71A.cir)



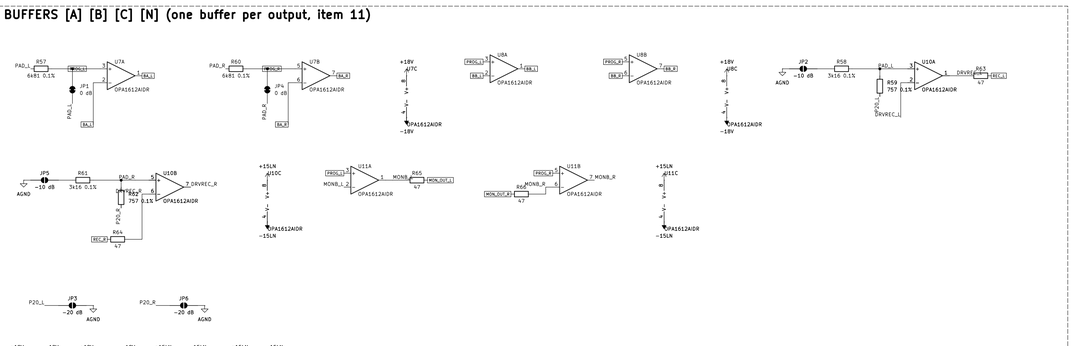
MONO [A3] (M/S matrix + VCA on S, mono−ms.cir; resistors NOTE)



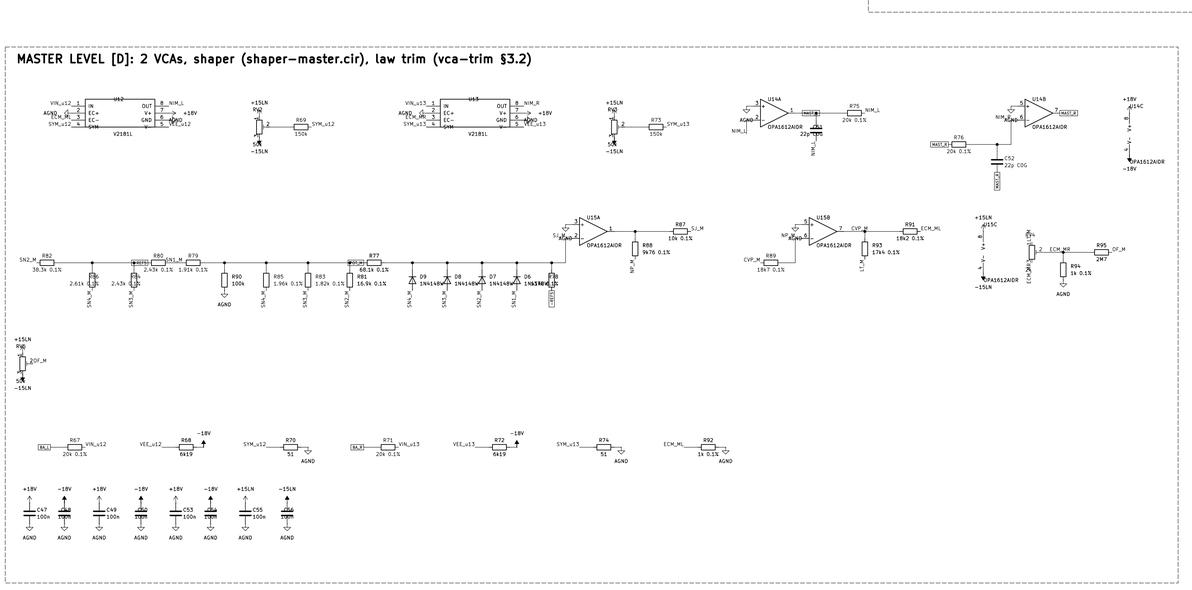
INS_SEND driver + INS_RET buffer (slots 9−10, JMS jumper on the backplane)



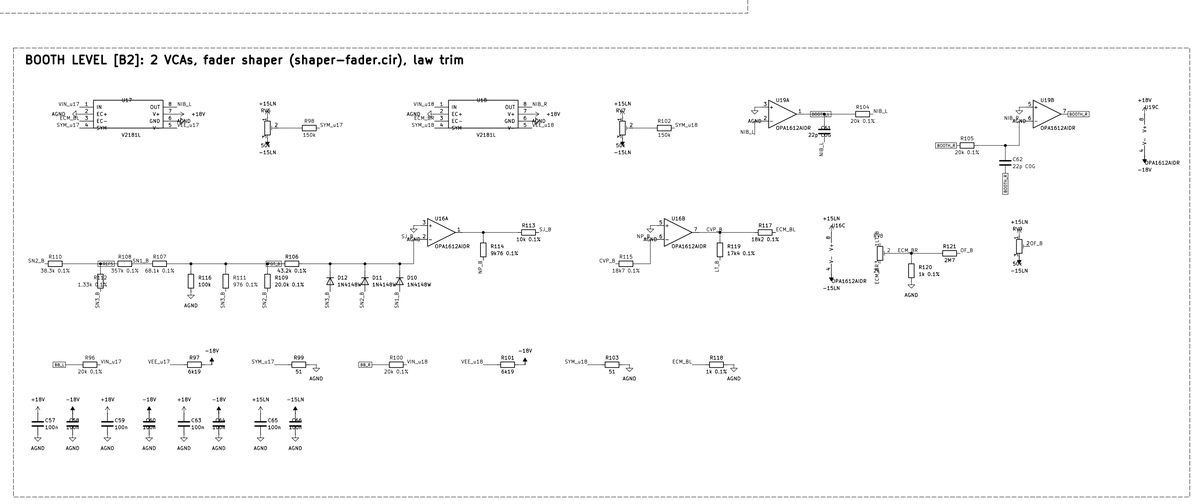
BUFFERS [A] [B] [C] [N] (one buffer per output, item 11)



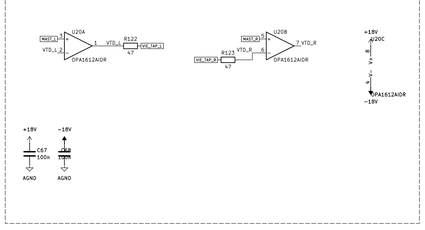
MASTER LEVEL [D]: 2 VCAs, shaper (shaper−master.cir), law trim (vca−trim §3.2)



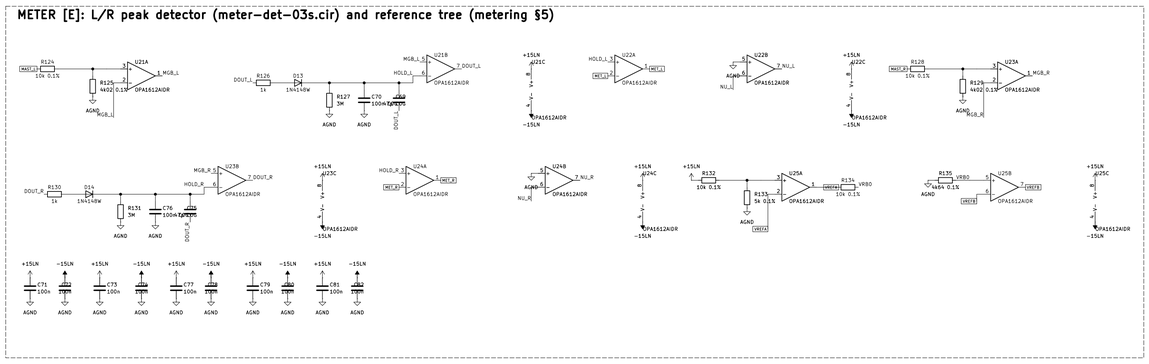
BOOTH LEVEL [B2]: 2 VCAs, fader shaper (shaper−fader.cir), law trim



VIE_TAP [V]: driver after MASTER LEVEL to b31−b32



METER [E]: L/R peak detector (meter−det−03s.cir) and reference tree (metering §5)



CONNECTORS: J2 to the OUTPUT board, J3 to the CONTROL board



PWR_FLAG (for the ERC check only)



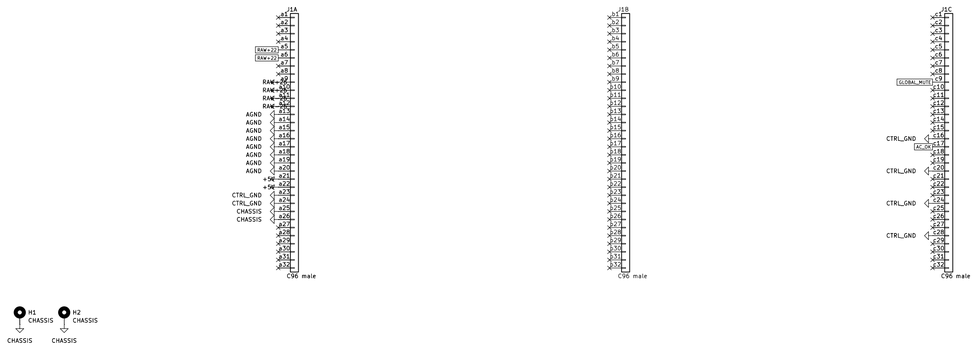
Generated by pcb/tools/legglbile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: master−segnale.kicad_sch

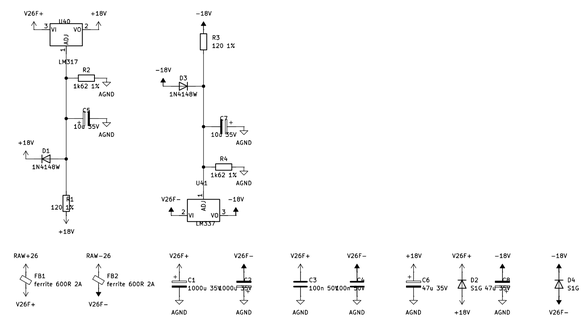
Title: maelcum MASTER signal board (dec. 140) − draft v0.1 − readable sch

Size: User	Date:
KiCad E.D.A. 10.0.7	Rev: 0.2−draft
	Id: 1/1

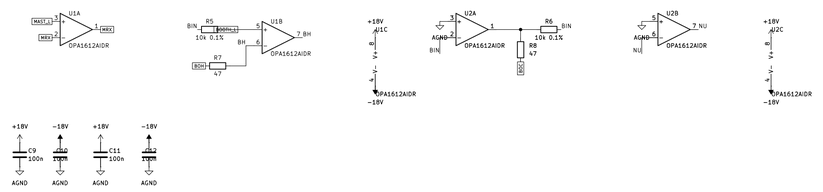
BACKPLANE C96 (MASTER-OUT class, dec. 143): power only



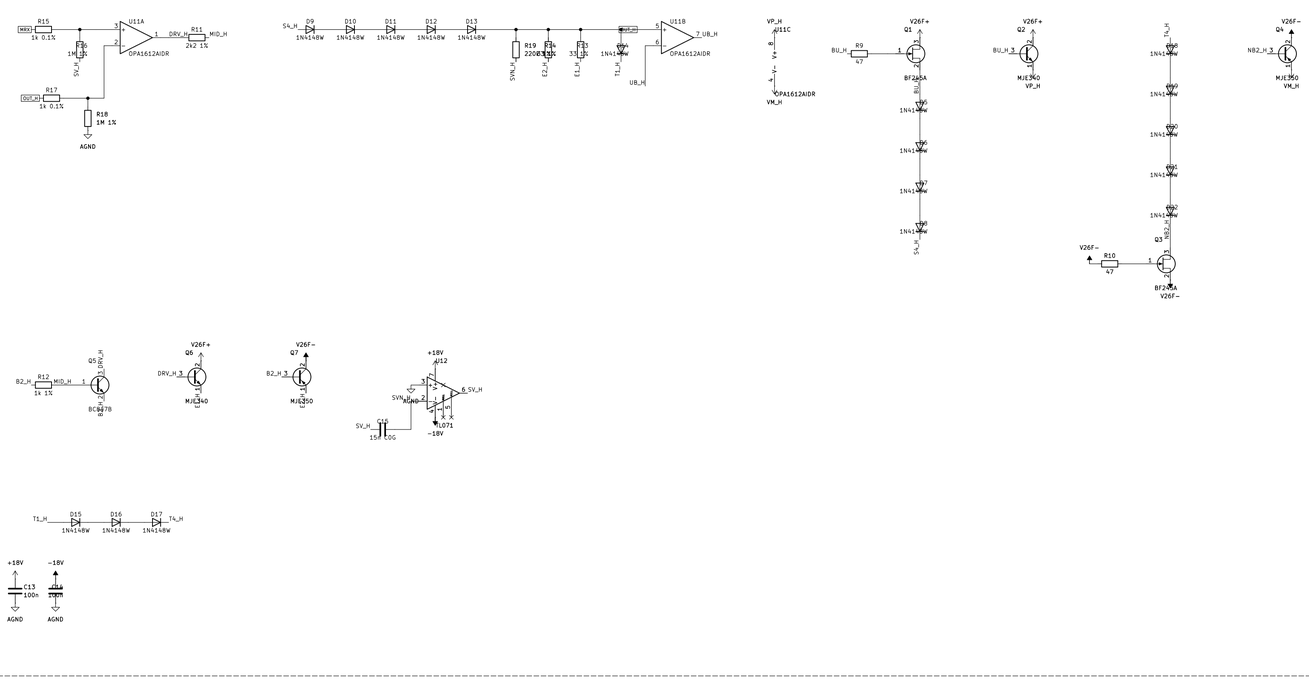
POWER [M]: filtered +-26 static rails, local +-18 (LM317/337)



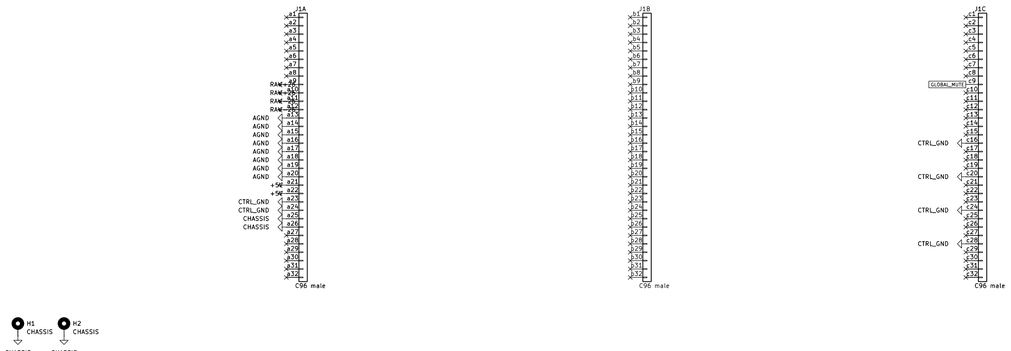
MASTER RECEIVER and BOOTH DRIVER [B3] (direct OPA1612, §3.4)



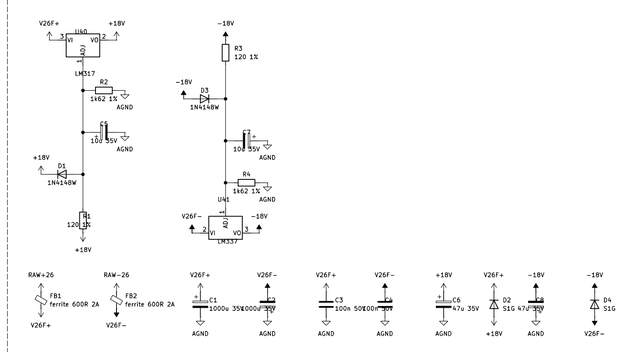
OUTPUT STAGE hot [F]+[G] (bootstrap, dec. 86)



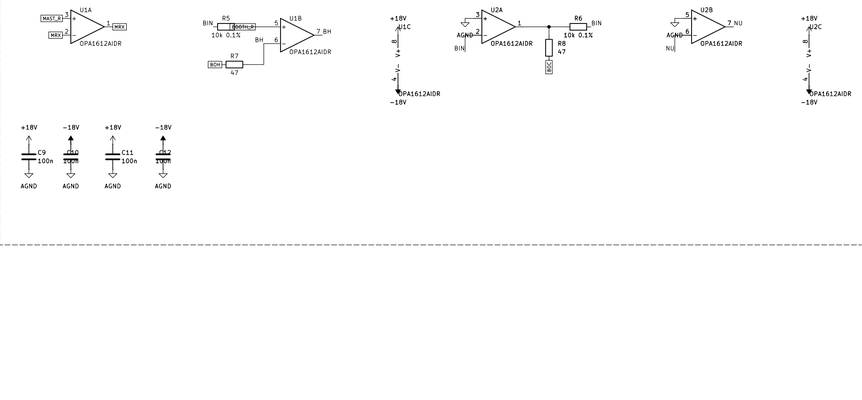
BACKPLANE C96 (MASTER-OUT class, dec. 143): power only



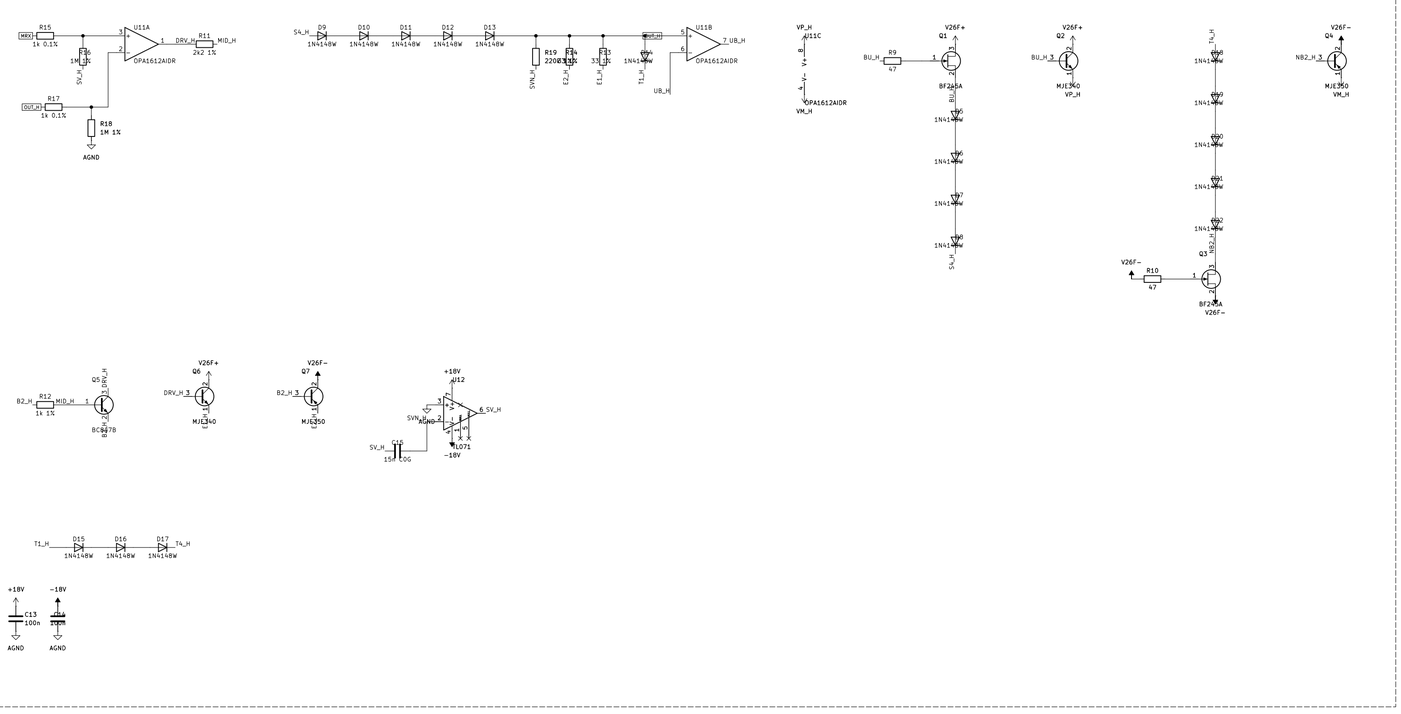
POWER [M]: filtered +-26 static rails, local +-18 (LM317/337)



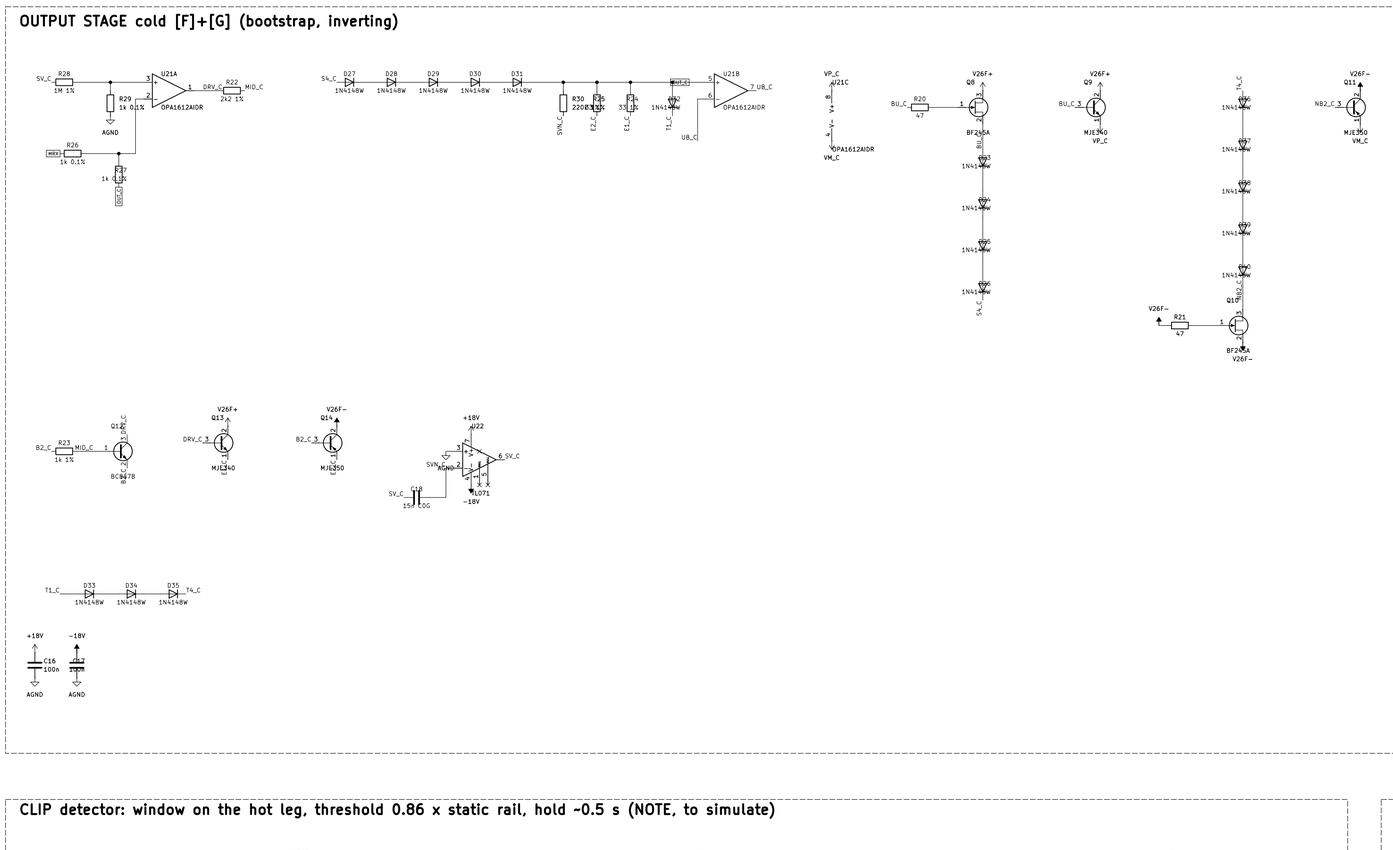
MASTER RECEIVER and BOOTH DRIVER [B3] (direct OPA1612, §3.4)



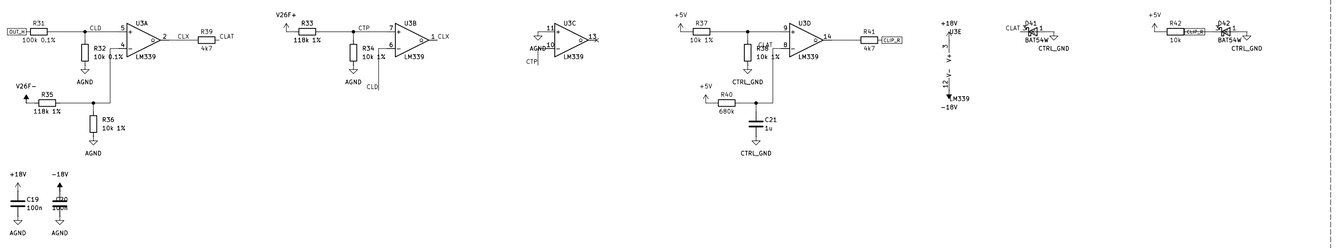
OUTPUT STAGE hot [F]+[G] (bootstrap, dec. 86)



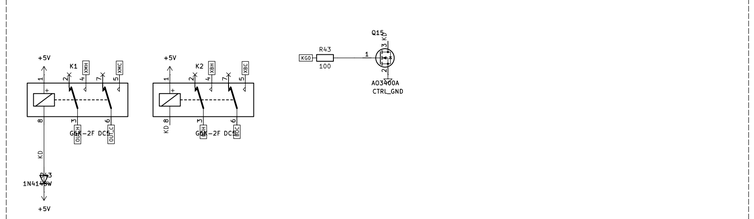
OUTPUT STAGE cold [F]+[G] (bootstrap, inverting)



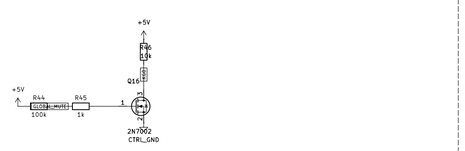
CLIP detector: window on the hot leg, threshold 0.86 x static rail, hold ~0.5 s (NOTE, to simulate)



MASTER and BOOTH RELAYS (G6K-2F, 5 V, 237 ohm; energised = sound, failsafe)



RELAY DRIVE from GLOBAL_MUTE (c9): low = sound



CONNECTORS: J2 from the signal board, J3 to the rear connector board



PWR_FLAG (for the ERC check only)



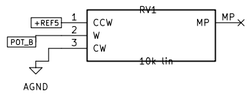
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

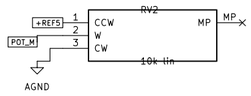
Sheet: /
File: master-uscita-R.kicad_sch

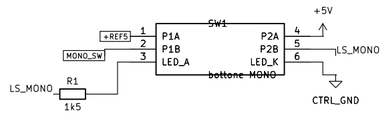
Title: maelcum MASTER OUTPUT board R (dec. 142-143) - draft v0.1 - rea

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

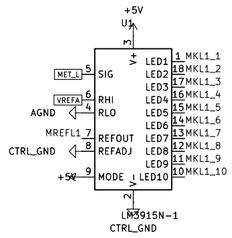
POTS and MONO BUTTON (positions from the panel render)

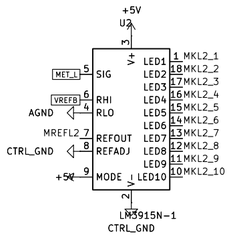


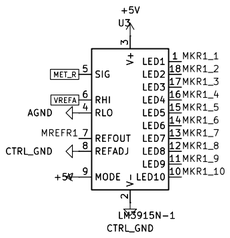


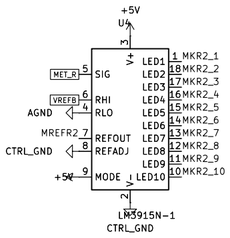


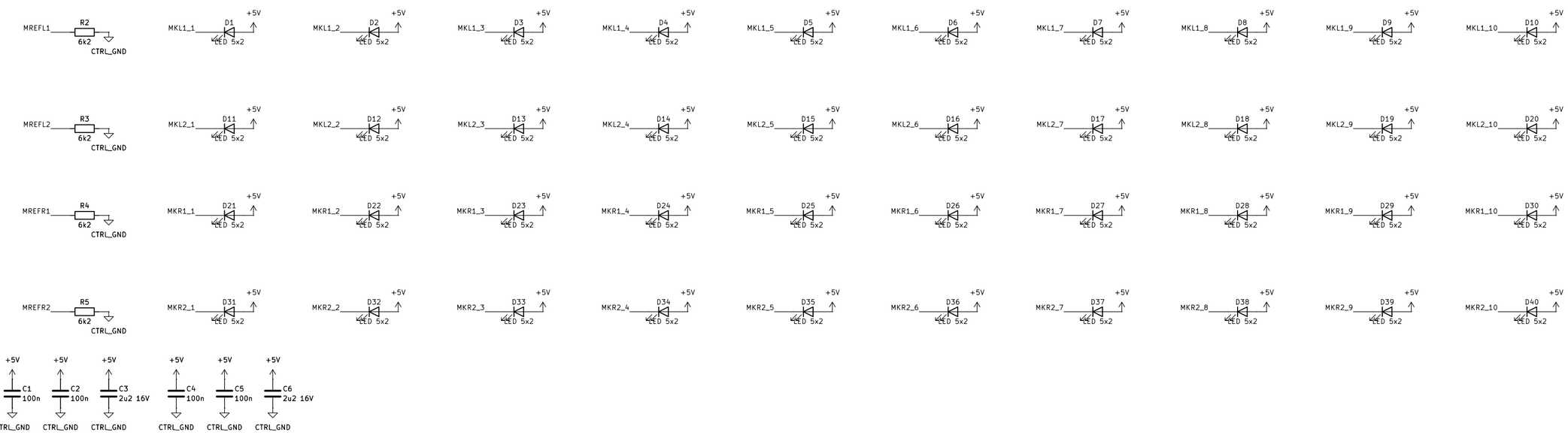
METERS L and R: 2 cascaded LM3915 per side, 20 LEDs



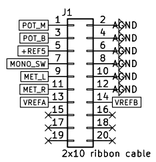


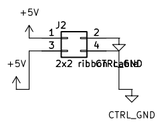




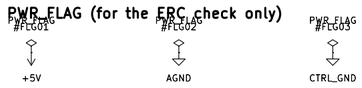


CABLES: J1 to the signal board (J3), J2 to the L output board (J4)





PWR_FLAG (for the ERC check only)



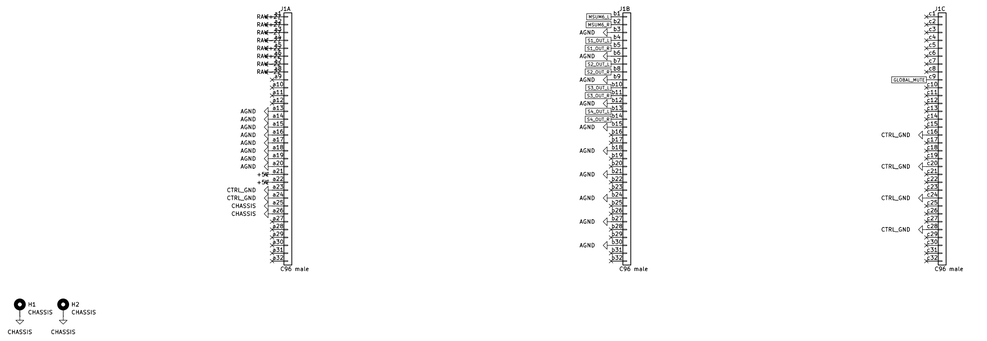
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: master-comandi.kicad_sch

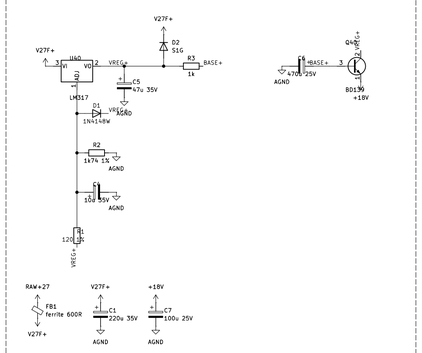
Title: maelcum MASTER control board — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1

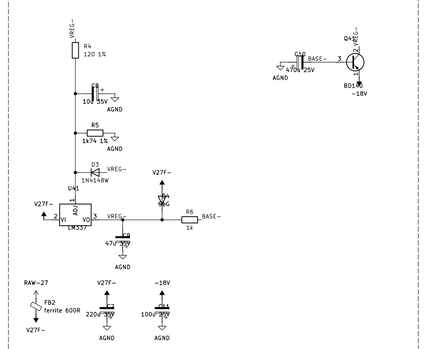
BACKPLANE C96 (MIC/GEN class, MSUM6)



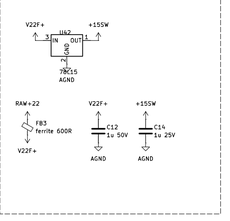
POWER: +18V (LM317 + BD139)



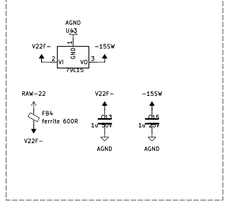
POWER: −18V (LM337 + BD140)



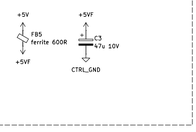
POWER: +15SW (78L15)



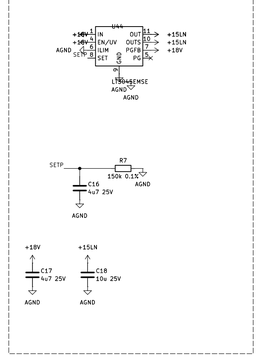
POWER: −15SW (79L15)



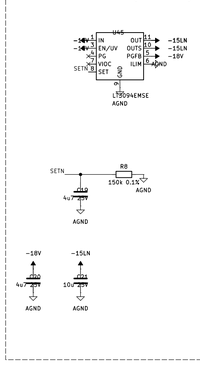
POWER: +5VF



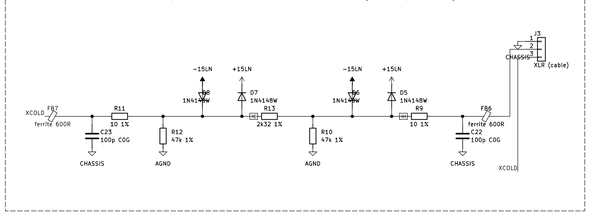
POWER: +15LN (LT3045)



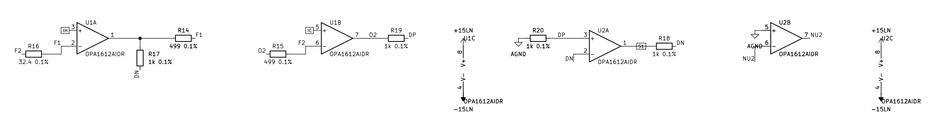
POWER: −15LN (LT3094)



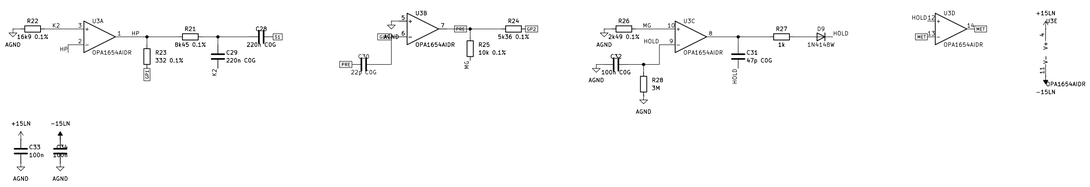
XLR INPUT: RFI, clamp, 2.32k load, 2x47k bias (mic−pre.inc)



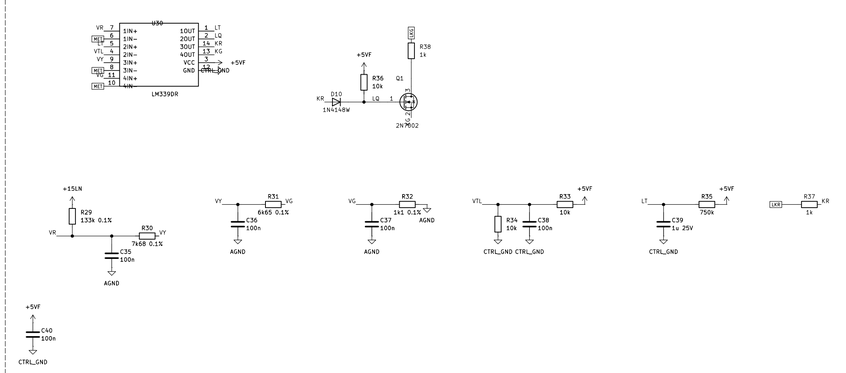
INA +30.05 dB and unity differential (OPA1612)



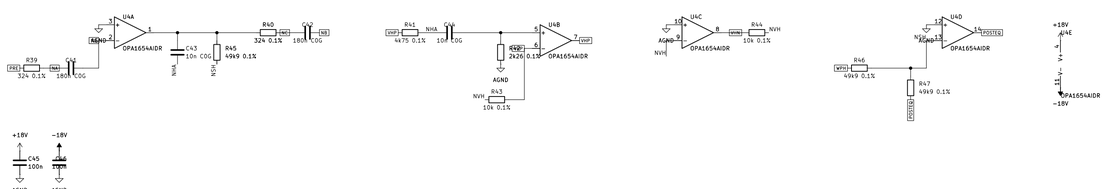
HIGH−PASS 60 Hz, STAGE 2 (0...+30 dB), LED DETECTOR (OPA1654)



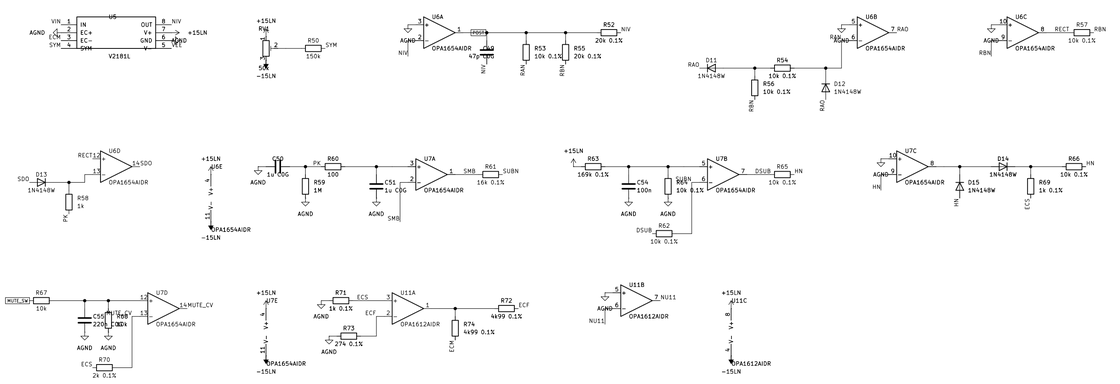
Three−colour LED (LM339)



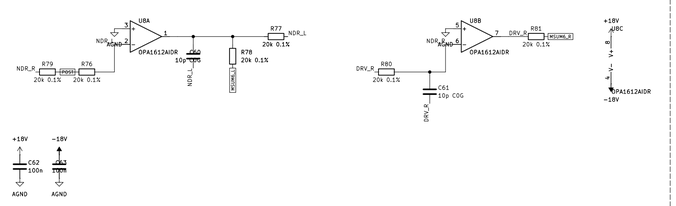
EQ2 (+−18)



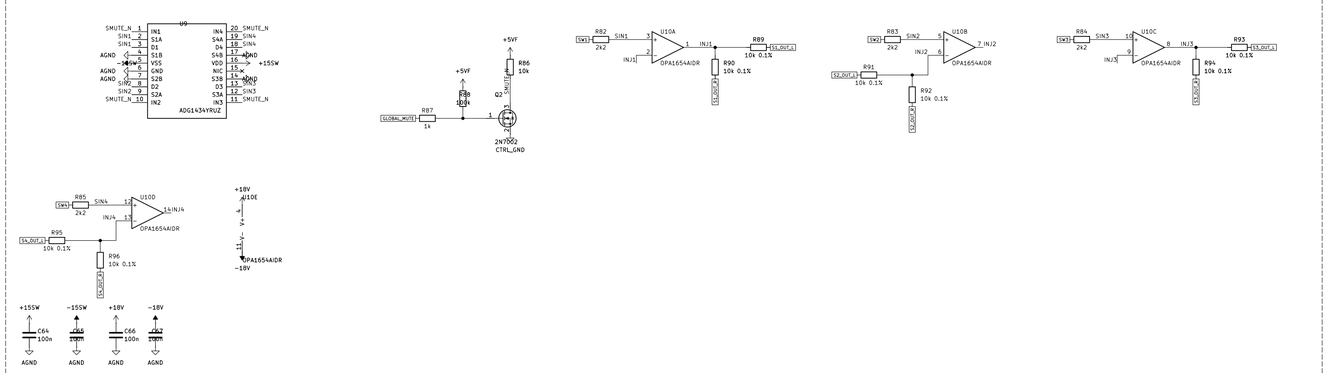
MUTE and limiter VCA (dec. 111, 145; sim/mic/mic−limiter.cir)



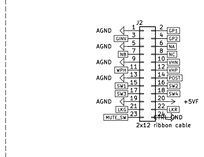
MSUM6 L/R DRIVERS (inverting, todo 227)



SENDS S1−S4 (pots on the cable, global mute, injection)



CONTROL BOARD CABLE (J2)



PWR_FLAG (for the ERC check only)



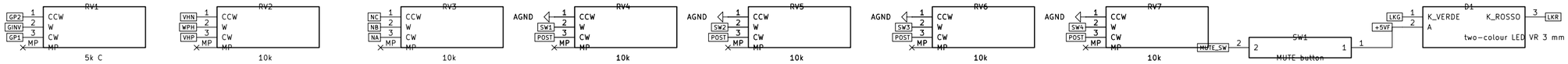
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mic.kicad_sch

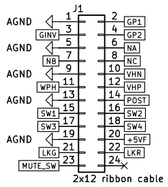
Title: maelcum MIC (dec. 97) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1−draft
KiCad E.D.A. 10.0.7		Id: 1/1

CONTROLS (positions from the panel render)



CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)

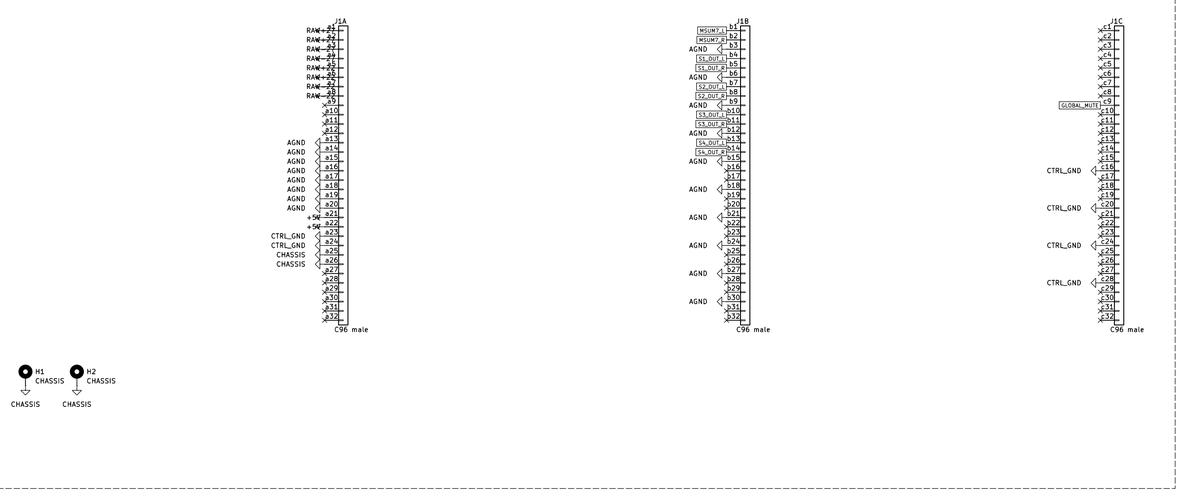
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mic-comandi.kicad_sch

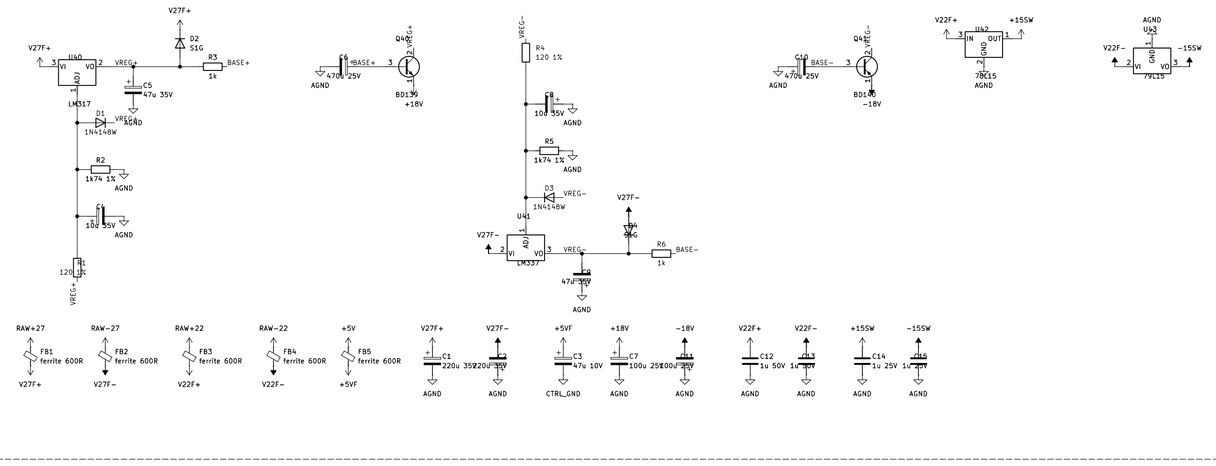
Title: maelcum MIC control board — draft v0.1 — readable schematic

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

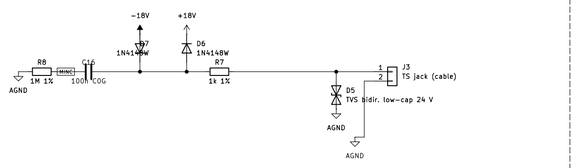
BACKPLANE C96 (MIC/GEN class, MSUM7)



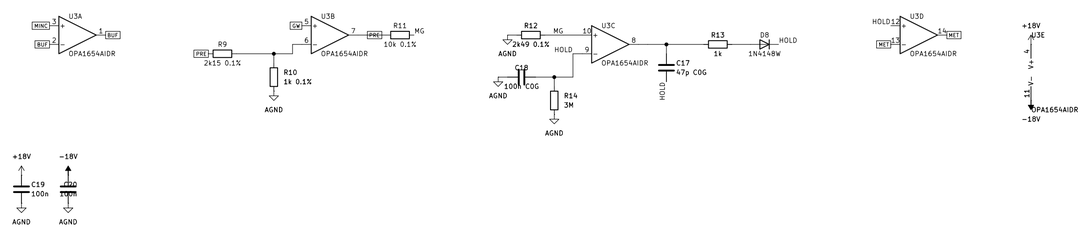
POWER +−18 audio, +−15 switch, +5



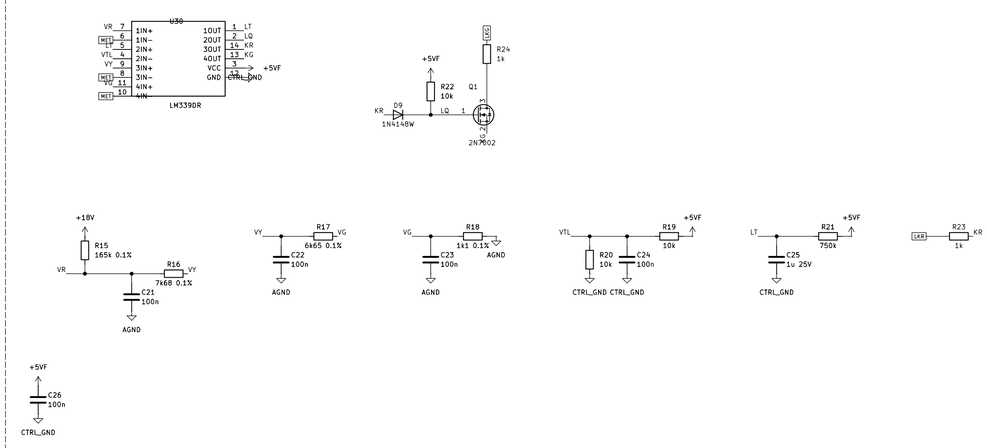
TS JACK INPUT: series R, clamp, TVS, film C, 1M bias



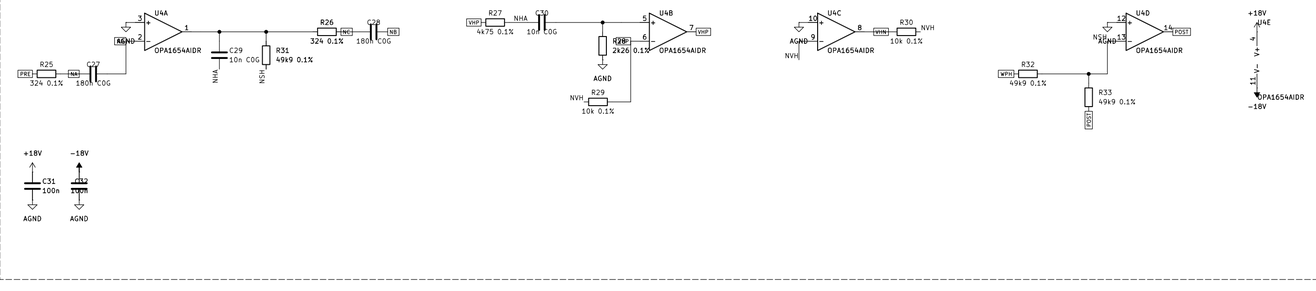
BUFFER, MAKE−UP +10 dB, LED DETECTOR (OPA1654)



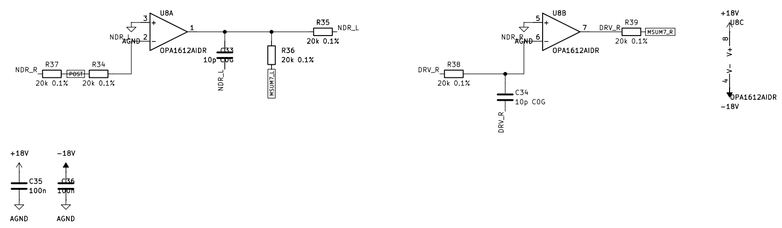
Three−colour LED (LM339)



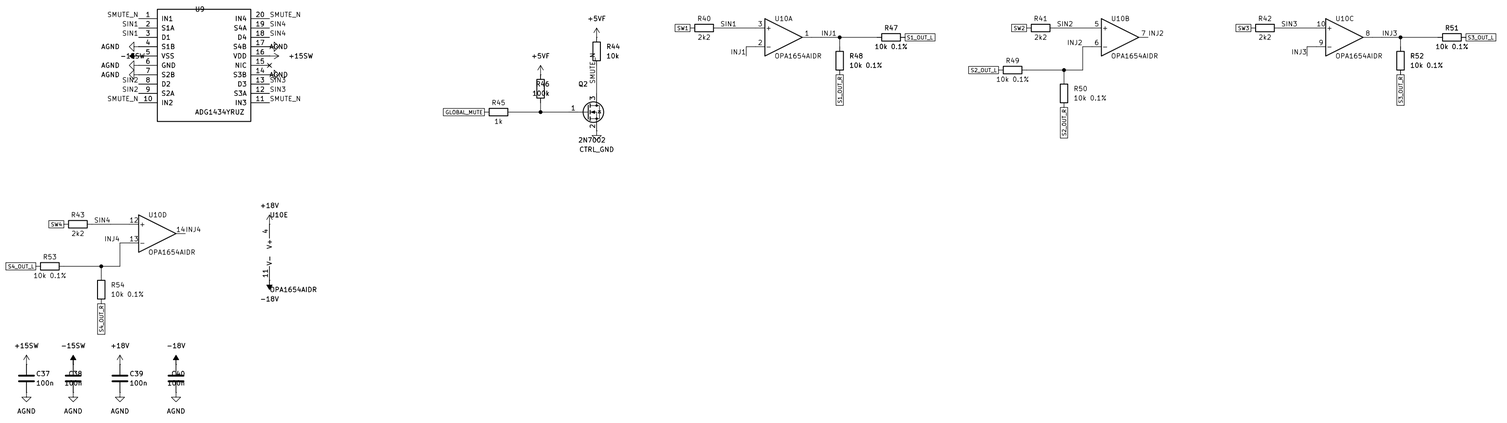
EQ2 (+−18)



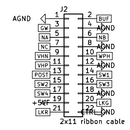
MSUM7 L/R DRIVERS (inverting, todo 227)



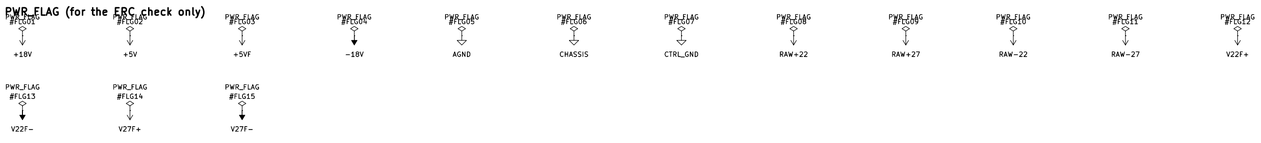
SENDS S1−S4 (pots on the cable, global mute, injection)



CONTROL BOARD CABLE (J2)



PWR_FLAG (for the ERC check only)



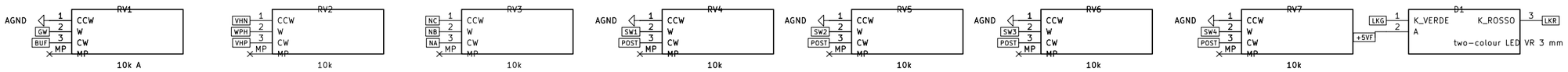
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mono.kicad_sch

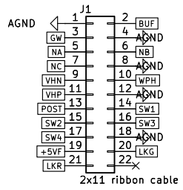
Title: maelcum MONO (dec. 97) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1−draft
KiCad E.D.A. 10.0.7		Id: 1/1

CONTROLS (positions from the panel render)



CABLE TO THE MAIN BOARD



PWR_FLAG (for the ERC check only)



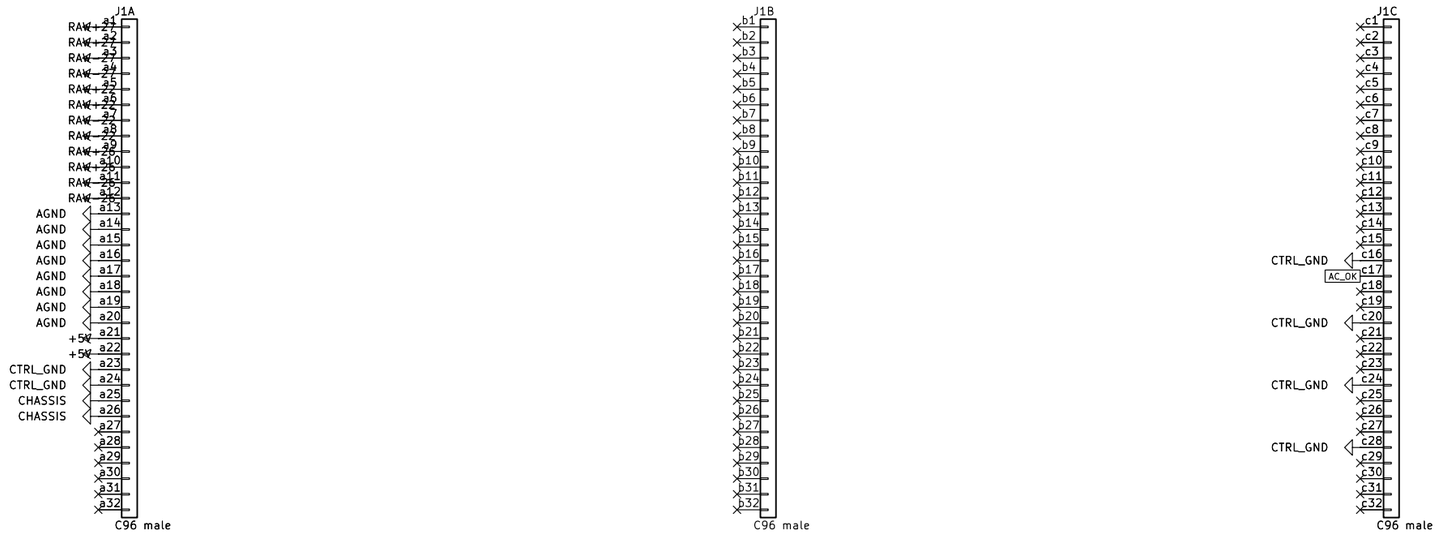
Generated by pcb/tools/leggibile.py from the same netlist as the PCB: do not edit by hand

Sheet: /
File: mono-comandi.kicad_sch

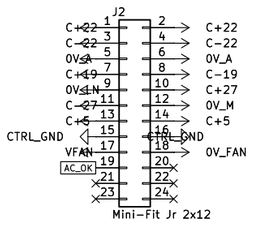
Title: maelcum MONO control board — draft v0.1 — readable schematic

Size: User	Date:	**Rev: 0.1-draft**
KiCad E.D.A. 10.0.7		Id: 1/1

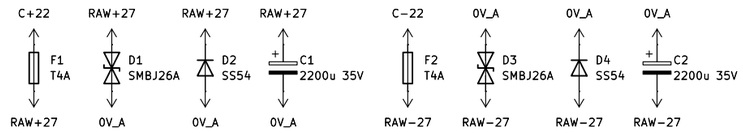
BACKPLANE C96 (PSU-IN slot): whole row a, AC_OK on c17



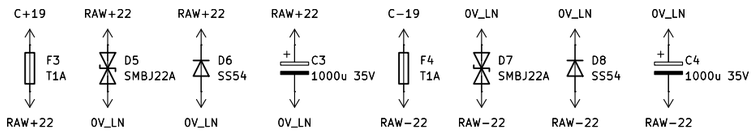
CABLE FROM THE BOX (20 conductors, §3)



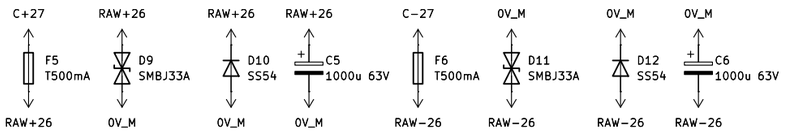
AUDIO RAIL +-22.0 V (raw +-27 pins): T4 A, TVS 26 V, 2.2 mF



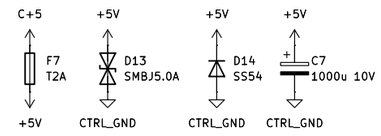
LN RAIL +-19 V (raw +-22 pins): T1 A, TVS 22 V, 1 mF



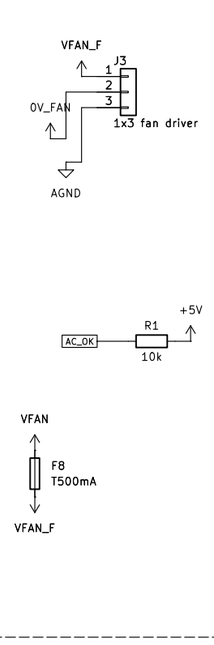
RAIL [M] raw +-27 V (raw +-26 pins): T500 mA, TVS 33 V, 1 mF



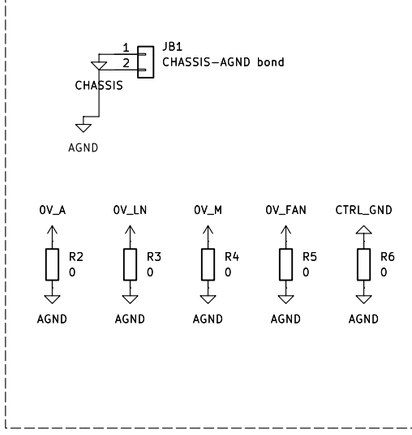
RAIL +5 V: T2 A, TVS 5.0 V, 1 mF



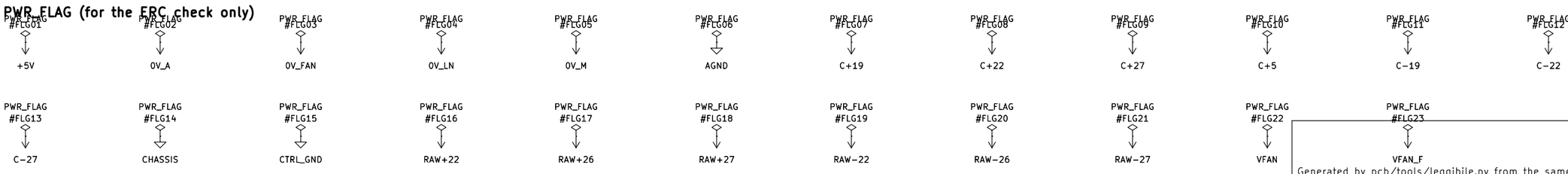
V_FAN (T500 mA) and AC_OK



analogue-0V STAR (§4.2, grounding §2)



PWR_FLAG (for the ERC check only)



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Sheet: /
File: psu-in.kicad_sch

Title: maelcum PSU-IN (dec. 113) — draft v0.1 — readable schematic

Size: User	Date:	Rev: 0.1-draft
KiCad E.D.A. 10.0.7		Id: 1/1